dealroom.co

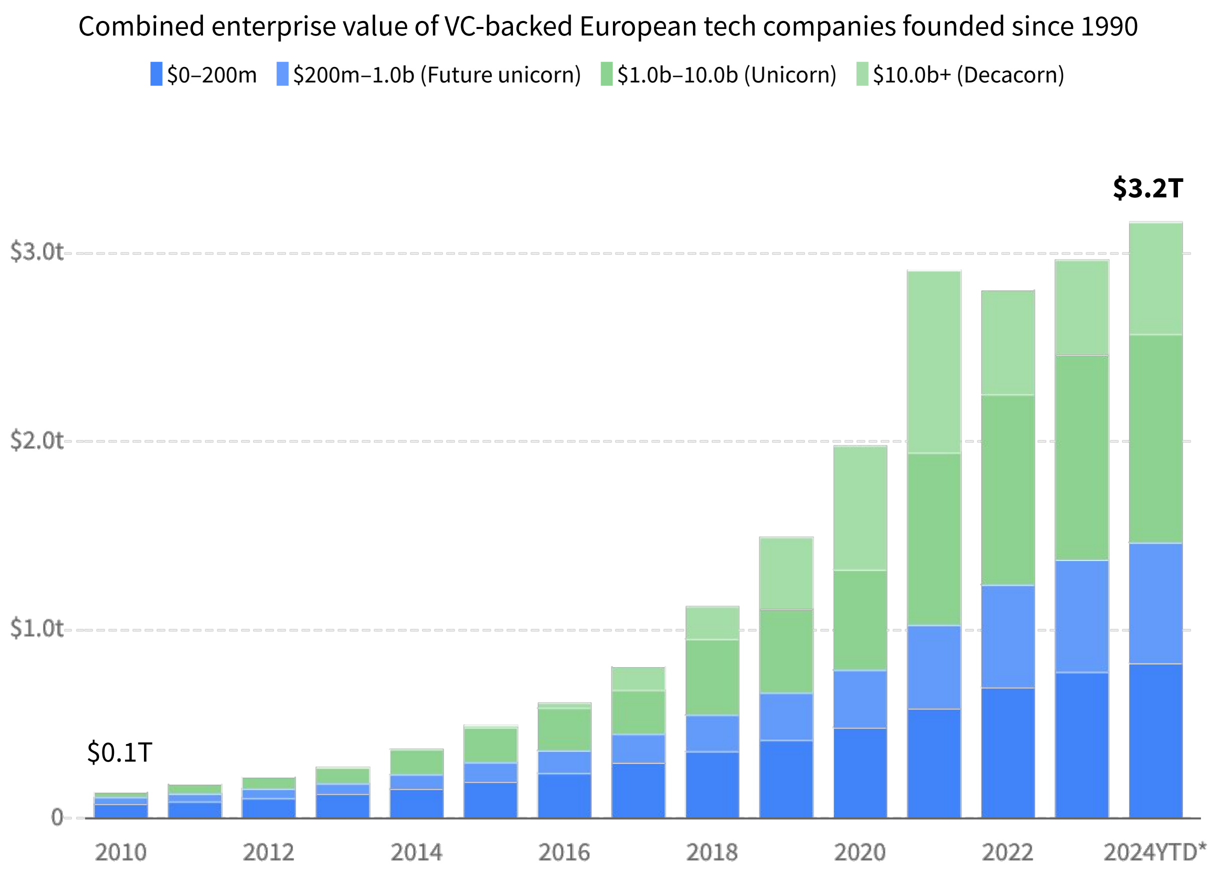
ENTROPY INDUSTRIAL CAPITAL

# Realtech: (Re)building Europe's critical industries

REALTECH CONFERENCE

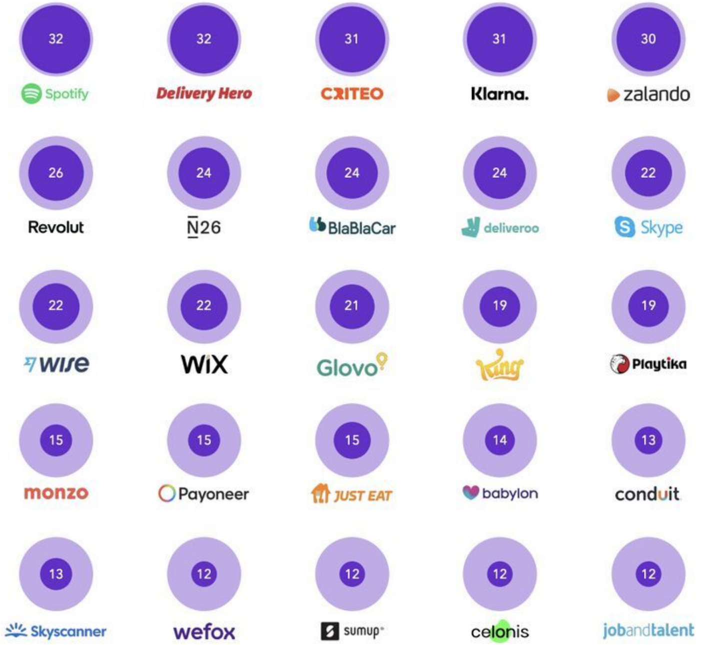
London, 2024

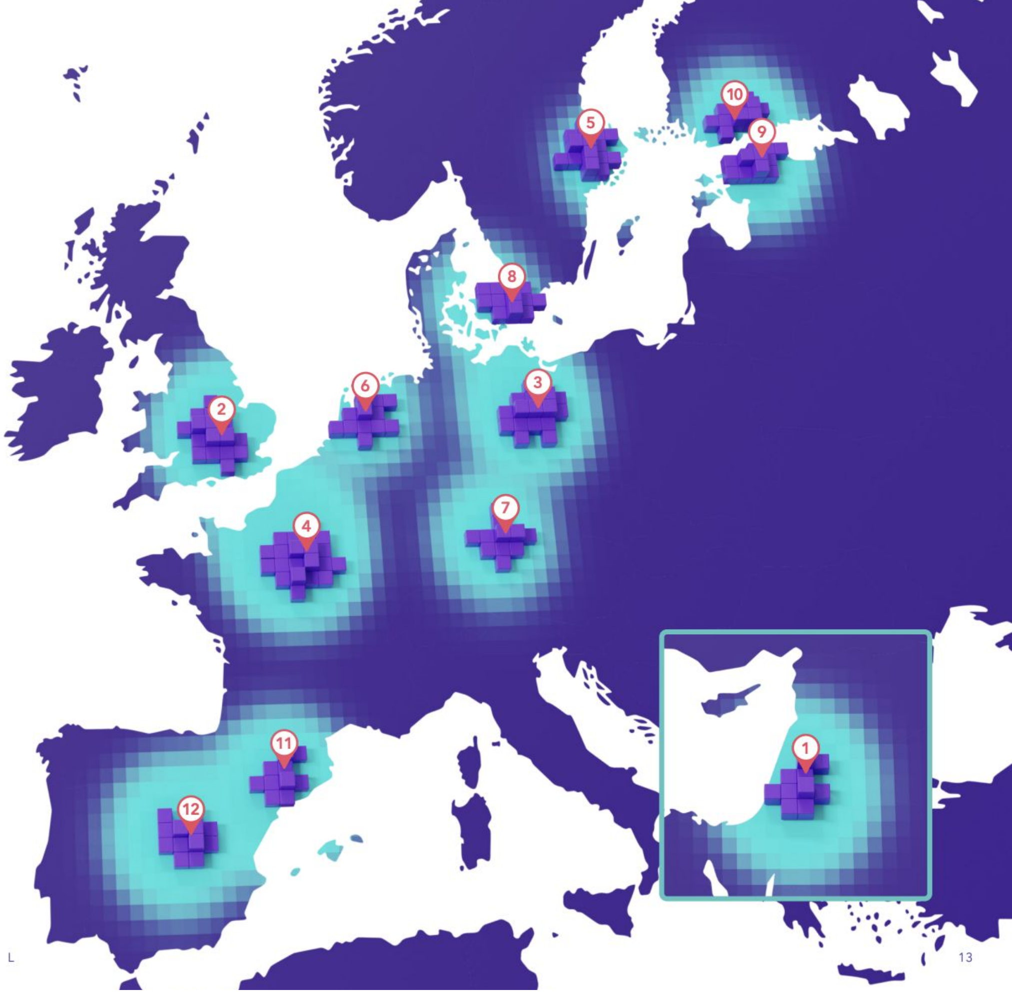
# Europe's tech ecosystem is now worth over $3.2 trillion.




Source: Dealroom.co

# More entrepreneurial talent than ever in Europe and Israel.




 Source: Dealroom.co & Accel

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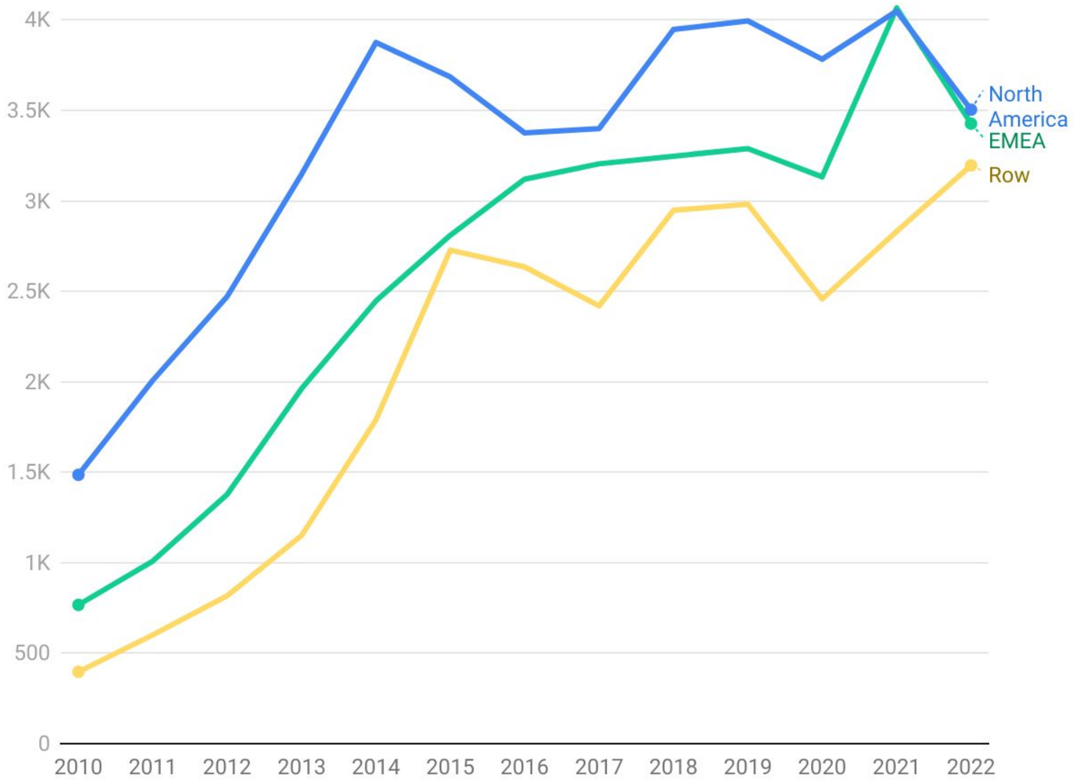
# Every year, about 3,500 new startups receive their first investment from a VC in EMEA.

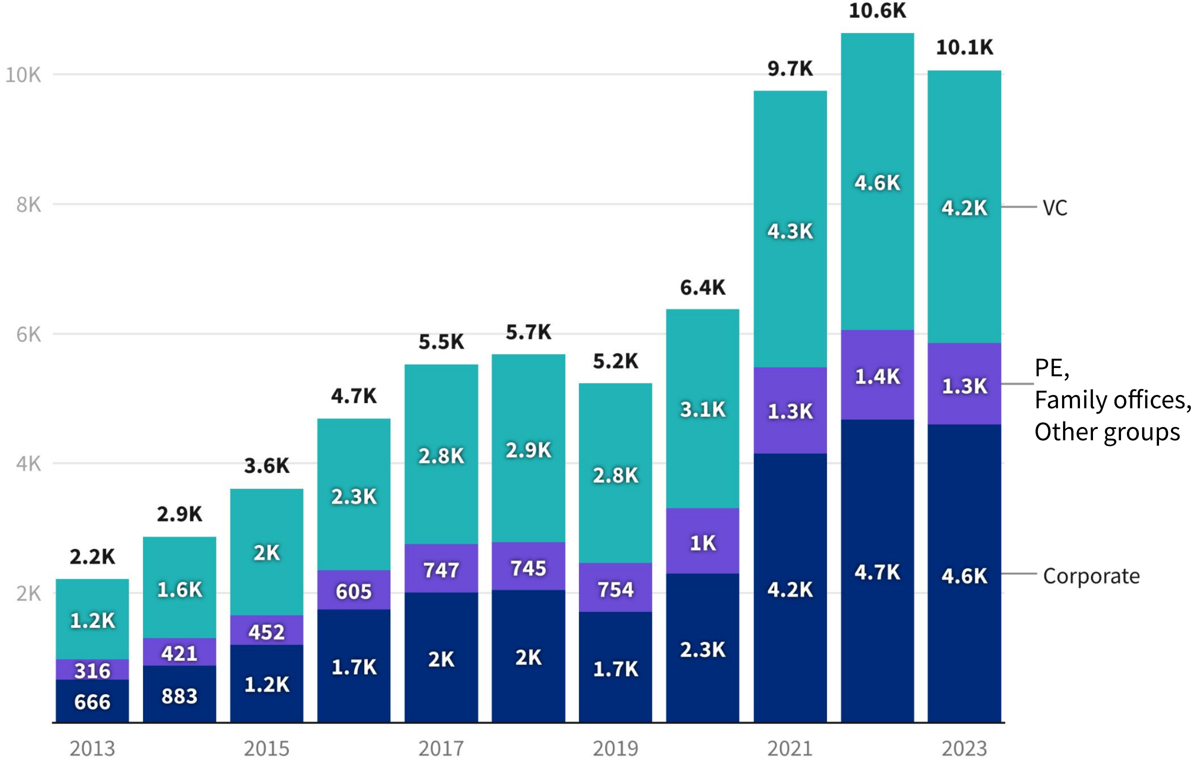
Number of first-time VC rounds per region




Source: Dealroom.co

# Investor interest in Europe has grown more than 4x and has stayed elevated even during 2023.

Unique number of investors in EMEA startups (at least one investment in the year)



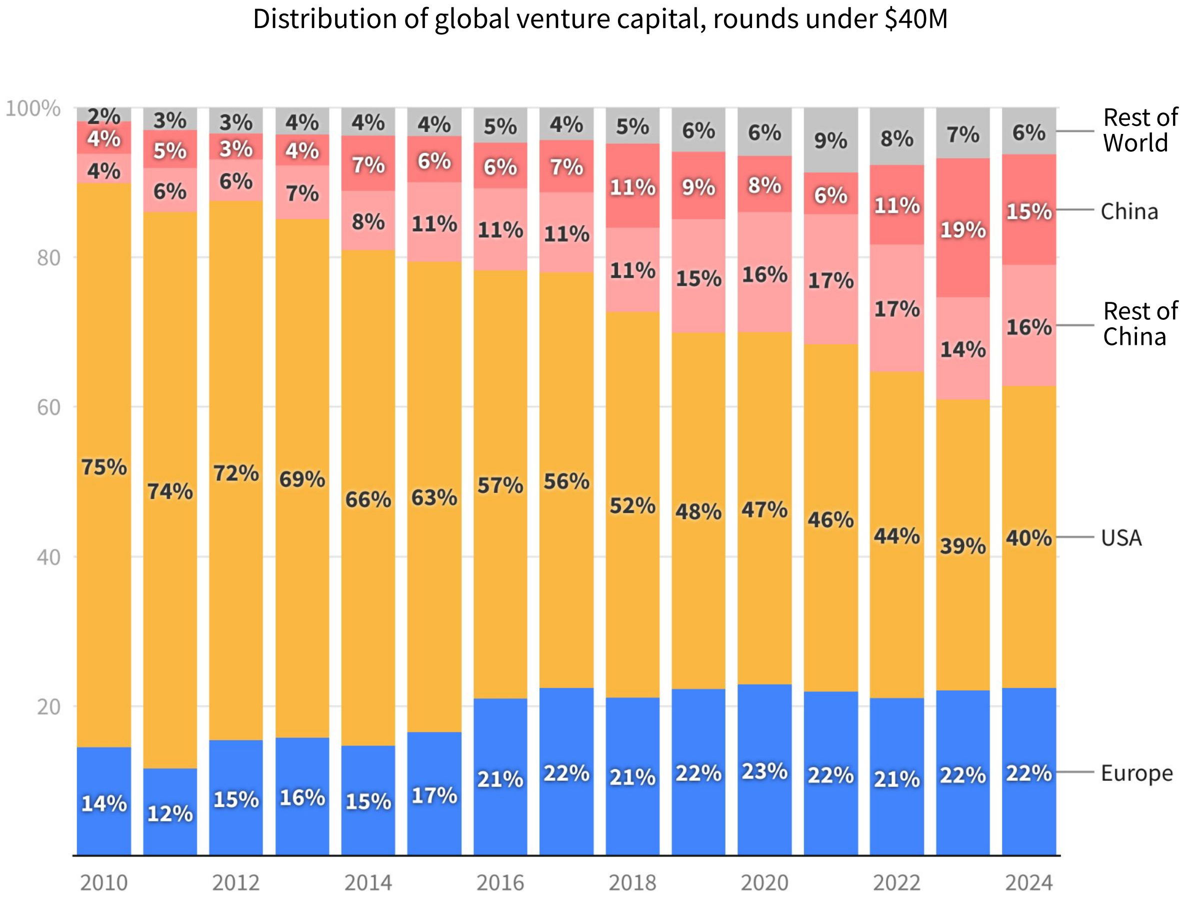

*A unique investor is one that did one or more investments.*

Source: Dealroom.co

# Europe's share of global early-stage VC is at a record 22% in 2024.

(<$40M rounds)

Double the level of 2011.






dealroom.co

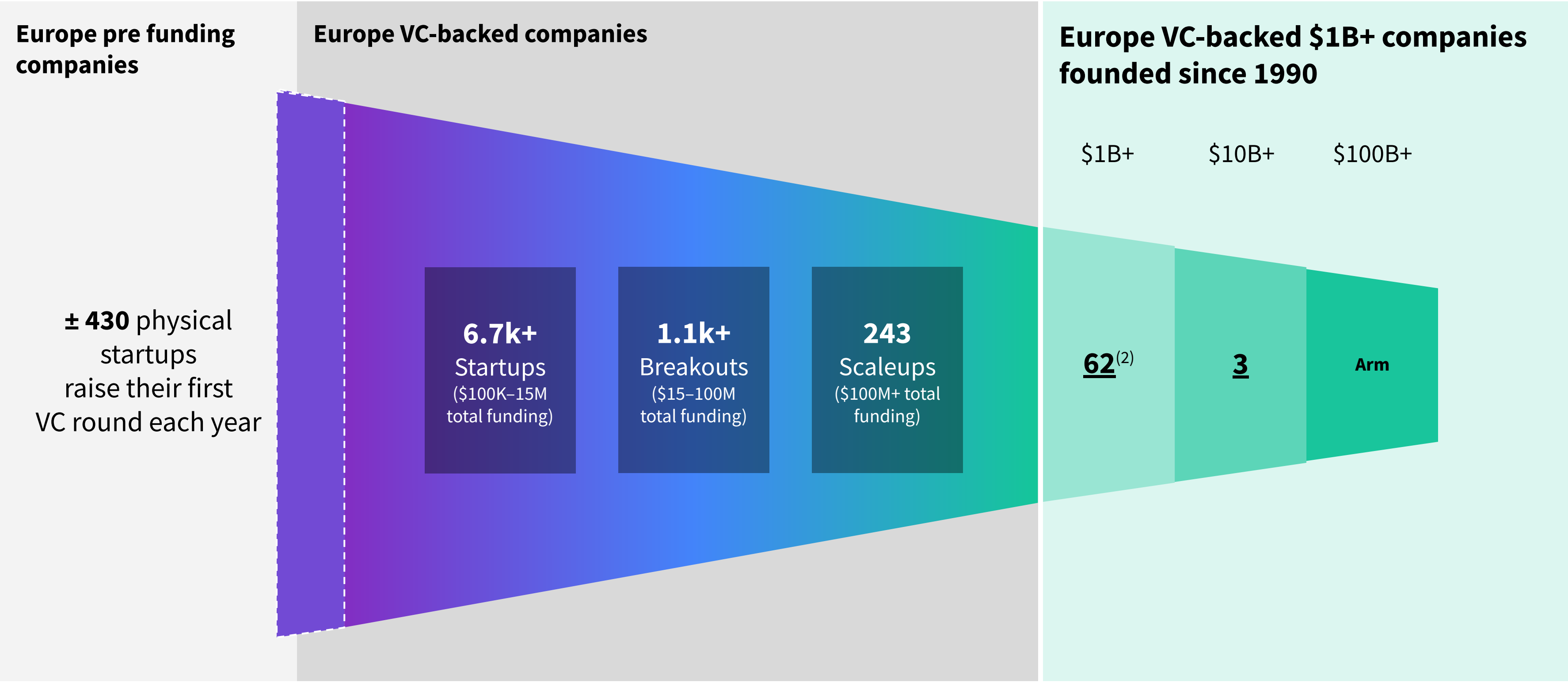
1 The rise of Europe

**2 Entering the new industrial age**

3 Where Europe stands

4 Unlocking European Industrial tech

# 15,000 VC-backed physical industry startups in Europe.



# Entering a new era in tech, more driven by Physical Tech.

## PC era

Selling hardware and software to large enterprises and homes

arm
ASML

## Internet era

Entire industries shifting online, to cloud, and to mobile. Big home market was huge advantage (Amazon, Facebook). In this second era, European players could go up against US counterparts

Spotify
Booking.com
adyen
Revolut

## Next 20 years

AI deeply embedded in physical world including biotech, healthcare, defense, manufacturing, et cetera

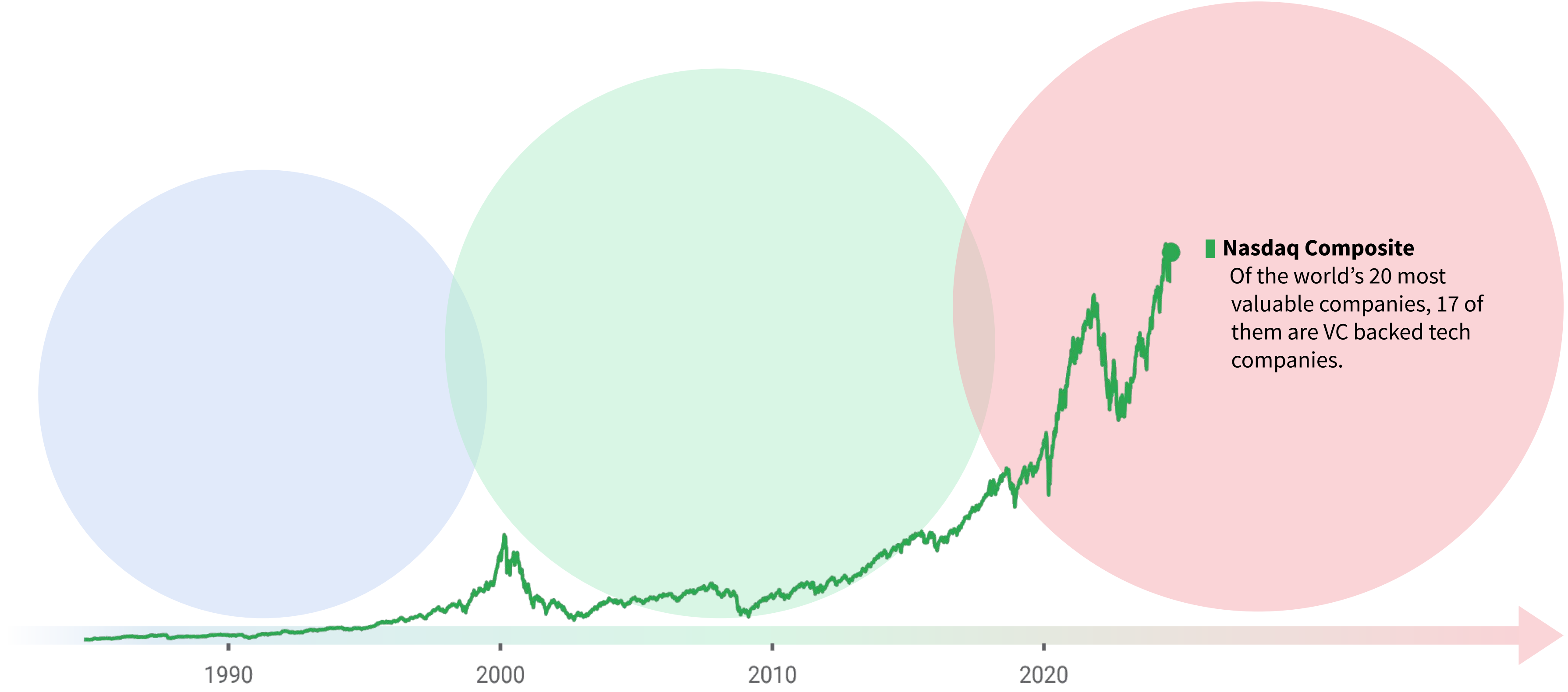
Helsing
northvolt
QUANTUM SYSTEMS
AGILE ROBOTS

1990 2000 2010 2020



Source: Dealroom.co.

# Each wave of tech has driven more value creation than the last.




 Source: Dealroom.co.

# 35% of the global economy is industrial; an output of $37 trillion (turnover >$80 trillion).

**Extraction**
mining $5T (4.5%)
+ agri $4T (4%) =
8.5% of GDP

**Industry**
manufacturing,
construction, utilities
**27% of GDP = $28 trillion**

**Enterprise**
finance, marketing, legal
consulting, logistics, etc.
**32% of GDP = $33 trillion**

**Consumer**
32% of GDP = $33 trillion





Source: Dealroom.co analysis. GDP data by World Bank. GDP split between B2B and B2C is indicative based on Eurostat data.

# Why Industrial/Physical Tech matters ...

## Climate

driving the green transition

## Geopolitical

ensuring freedom & sovereignty

## Scarcity

optimizing raw material use

## Skills gap

addressing workforce shortages

# Why it matters for Europe: a Critical Juncture

## Key Challenges

**Automotive Industry Decline**
Europe’s car industry, once dominant, is being outpaced by Tesla and Chinese EV makers, losing market share.

**Energy & Material Dependency**
High energy costs and reliance on imported fossil fuels & materials are straining competitiveness.

**Labor Shortages**
An aging workforce and slow digital adoption hinder innovation.

**Regulatory Pressure**
Digital regulations complicating operations across member states.

## Opportunities

**Green Transition**
Europe can lead in sustainable manufacturing through green tech and renewable energy.

**Industry 4.0**
Automation and digitalization offer a path to improved efficiency and competitiveness.

**Reshoring**
Regionalizing supply chains could boost resilience and local economies.

**Diverse Sectors**
Aerospace and machinery remain strong and ripe for innovation.

NEWS > FINANCIAL SERVICES

# Draghi demands €800B cash boost to stem Europe's rapid decline

"Without action, we will have to either compromise our welfare, our environment or our freedom," former ECB chief says in hotly anticipated report. Read the full paper here.



Source: https://www.politico.eu/article/mario-draghi-report-says-eu-must-spend-twice-as-much-after-wwii/

# Industrial tech / Physical Industries
overview

**Robots & Autonomous technology**
- Drones
- Industrial Robotics
- Autonomous vehicles
- Surgical robots
- General-purpose robotics intelligence

**Transportation**
- Green shipping
- Sustainable Aviation
- Trucks
- Electric vehicles

**Energy**
- Nuclear Energy
- Renewable energy generation
- Grid technology
- Energy storage
- Green hydrogen

**Quantum & Semiconductors**
- Quantum cryptography & communication
- Quantum computing hardware, quantum computing software, quantum sensing
- Chips manufacturing
- Radar, lidar & other advanced sensing

**Advanced materials**
- Advanced composites
- Smart textiles
- Multifunctional coatings
- Nanomaterials
- Bio-manufacturing

**Architecture, engineering and construction**
- Design software
- Construction robotics
- Sustainable building materials (green cement & steel)

**Space & Defence**
- Launch rockets
- In orbit servicing
- Deep space exploration
- Satellite communications
- Satellite imagery
- Space situational awareness

**Supply chain**
- Manufacturing tech
- Logistics
- Procurement
- Inventory management
- Freight & shipping

**Chemicals**
- Green fuels (e-fuels, green methanol)

**Critical minerals**
- Direct lithium extraction
- Battery recycling



Source: Dealroom.co Segments highlighted are not exhaustive but only explicative


***"Today's challenges demand real, tangible solutions - physical tech at the intersection of hardware and software. Robotics and automation will reduce the strain on our society and fast-track sustainable progress. Yes, regulations are slow, but in some cases, they are the most modern in the world. Just look at commercial drone applications: flying over densely populated areas is allowed and proven.***

"There has never been a better time to build physical ventures. Europe, with its top-tier talent, forward-thinking regulations, and strong industrial foundation, has everything needed to lead this shift and create a competitive edge over the world. "Made in Europe" is exportable again - we simply need to execute!

"Urban Ray is part of this movement, pushing logistics into a new dimension by making it airborne and automated, driving the future of urban transportation."

**Cem Uyanik**
CEO & Co-Founder at
Urban Ray

URBAN RAY

# European Upstream Space startups

» Explore the landscape

### Materials for space
Combined funding $ 168M

Airborne, PETROCERAMIC…, Fureho AB, NAWAH, Swegan, Surrey NanoSyst, Vulkam, Seram Coatings, PlasmaBound, ÉireComposites, Nanoker Researc

### Semiconductors for space
Combined funding $ 143M

Solar MEMS Tech, MAGICS Technolo, Paragraf, ID Quantique, Pilot Photonics, Ubotica Technol, MBRYONICS, Ligentec, Evince, Arquimea Ingeni, Emxys, AirMems, Saphyrion Sagl, ICsense

### Propulsion systems
Combined funding $ 44M

ThrustMe, Enpulsion, Gama Space, Pangea Aerospac, ION-X, Aurora Propulsi, Magdrive, Ienai SPACE, Pulsar Fusion, Orbital Machine, Neutron Star Sy

### Spacecraft parts, structures and payloads
Combined funding $ 122M

Mynaric, Oxford Space Sy, VEOWARE, Thorium, Satlantis, SWISSto12, AAC Clyde Space, ANYWAVES, Osol, CreoTECH, SpaceTech, Picosats, First light ima, ALMATECH, DAS Photonics, TTI NORTE, Lambda-X, Hee photonic la, Space Structure, Dynamic Optics, DCUBED, Golbriak Space, Mission Space L, Space Composite

### Cybersecurity for space missions
Combined funding $ 106M

Qascom, CYSEC, Arqit, Secure-IC, Crypta Labs

### In-space research
Combined funding $ 6.8M

SpacePharma, Yuri, Blue Skies Spac

### Mission planning and control
Combined funding $ 34M

Taitus, Spaceit oü, Aldoria, Neuraspace, Leanspace, OKAPI:Orbits, SPACEABLE, Aiko Space, Mission Space, Ephemersys, Exobotics UK Lt

### Spacecraft servicing; space debris removal and recycling
Combined funding $ 35M

ClearSpace, Future Space In, Kurs

### Earth observation satellites
Combined funding $ 670M

ICEYE, SatRev, Aerospacelab, Unseenlabs, Satellite Vu, OroraTech, Kuva Space, Sen Corporation, Kleos Space, Aistech Space, Karten Space, Hypernova Space

### Communication / connectivity satellites
Combined funding $ 4.7B

Eutelsat OneWeb, Kineis, E-Space, Hiber, Sateliot, Astrocast, OQ Tech, Methera global

### Satellite navigation
Combined funding $ 14M

Geoflex, Spaceopal, SYNTONY GNSS

### Ground infrastructure
Combined funding $ 261M

Arralis, ALL.SPACE, Alcan Systems, CONTEC, Leaf Space, GroundCom, KSAT GLOBAL, Homeport

### Launch vehicles
Combined funding $ 807M

PLD Space, Orbex Space, Isar Aerospace, Skyrora, Latitude, HyImpulse Techn, Sidereus Space, HyPrSpace

### In-space transportation spacecraft
Combined funding $ 202M

Lúnasa, D-Orbit, Exolaunch, SPACEDRONE, Exotrail, SmallSpark Spac, porkchop, UARX Space

### Stratospheric balloons and platforms
Combined funding $ 7.3M

Zephalto, EOS-X SPACESHIP, Zero 2 Infinity, B2Space

### Spaceplanes and hypersonic flight
Combined funding $ 90M

Destinus, Dark, Dawn Aerospace, Orbit Boy

### Space resource exploration
Combined funding $ 132M

PTScientists, Asteroid Mining, The Exploration, Team Tumblew…, Puli Space Tech, Space Mining Te, Regolithix, Metalysis, SpaceBIT

### Space utilities
Combined funding $ 50K

Maana Electric, Plus Ultra, Extraterrestria, indicium.space, The Lunar Grid

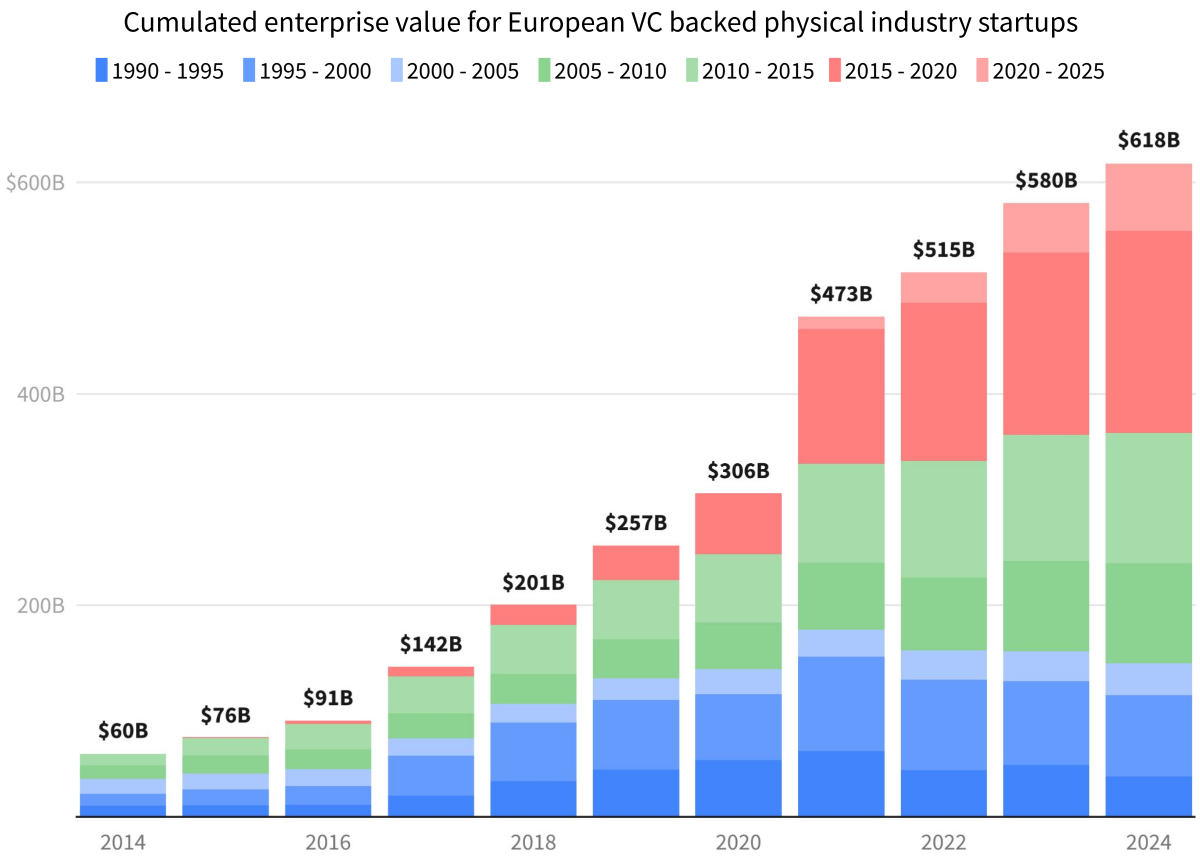
### In-space manufacturing
Combined funding $ 40M

CisLunar Indust, Space Forge, Moonfibre, Space Cargo Unl, ATLANT 3D

### In-space human presence
Combined funding $ 94M

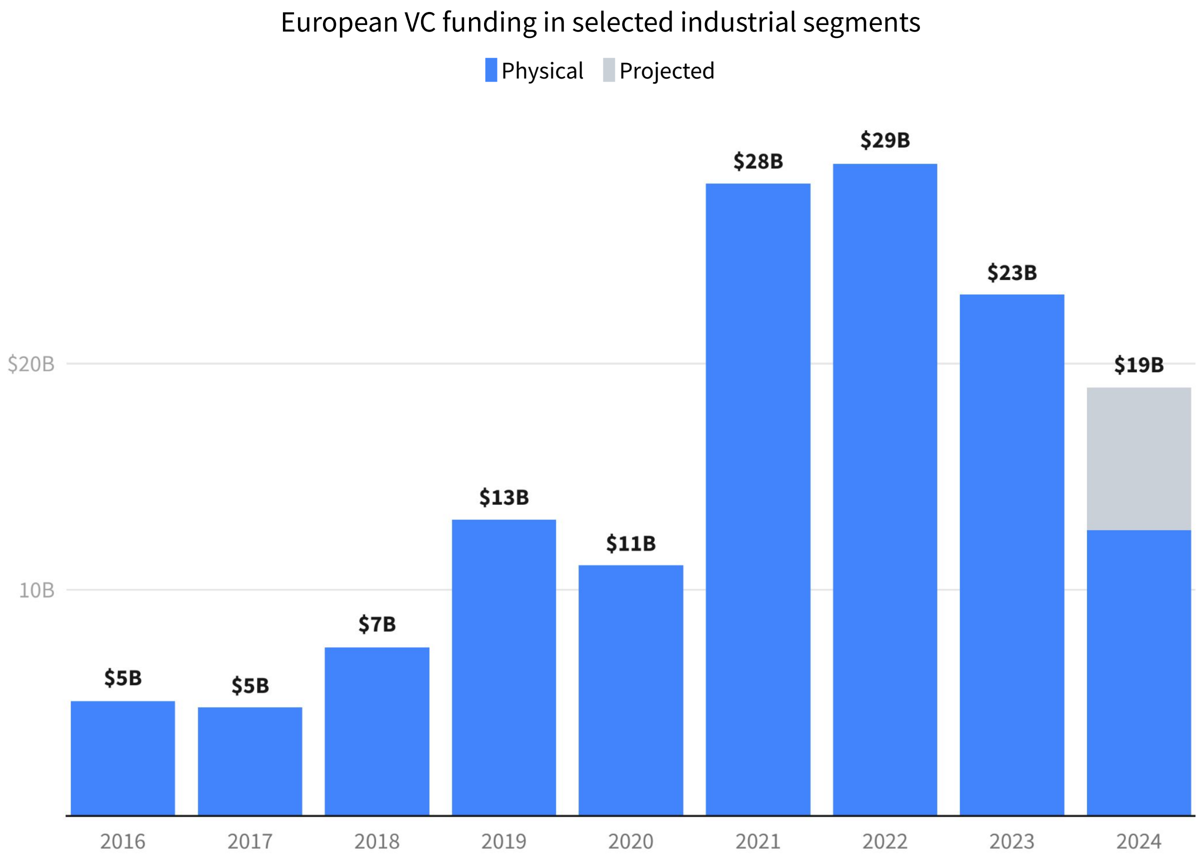
InterstellarLab, Solar Foods, Explaneta Space, SAGA Space Arch, Spartan Space

# European VC-backed industrial tech startups have passed $600B in combined value.




Source: Entropy Capital, Dealroom.co
Data as of Sep 12th 2024.

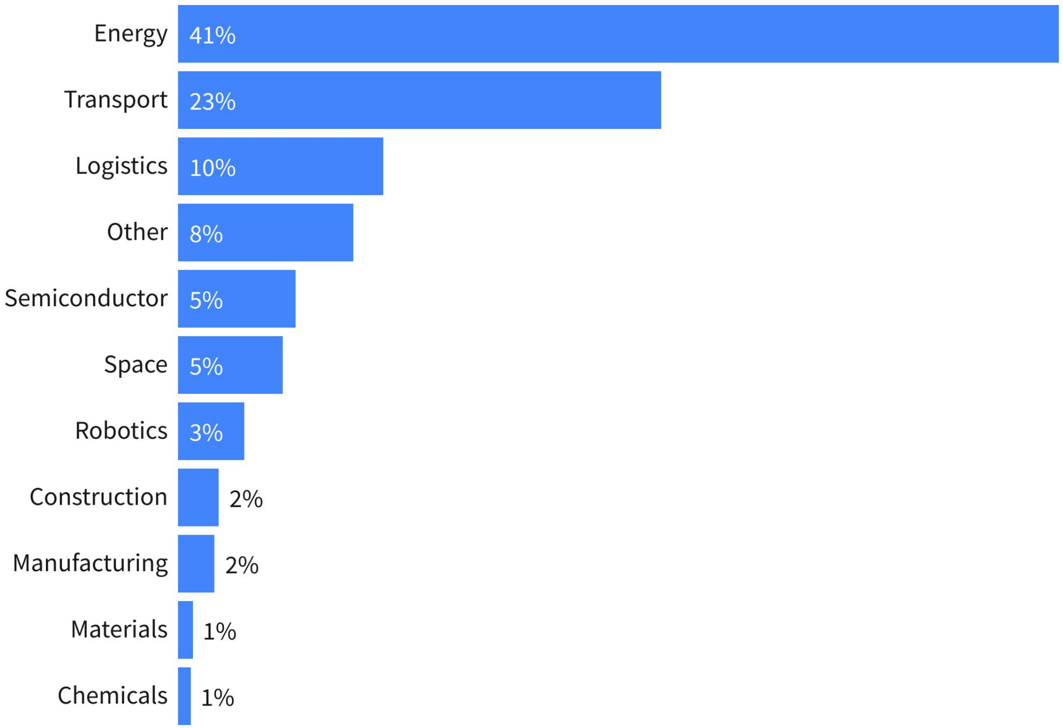
# VC funding in industrial tech in Europe is expected to hit $19B this year.




Source: Entropy Capital, Dealroom.co
Data as of Aug 29th 2024.

# Energy is the leading sector - raised 41% of all industrial tech VC in Europe.

Share of VC funding allocated per physical industries in Europe (2016 - 2024 YTD)

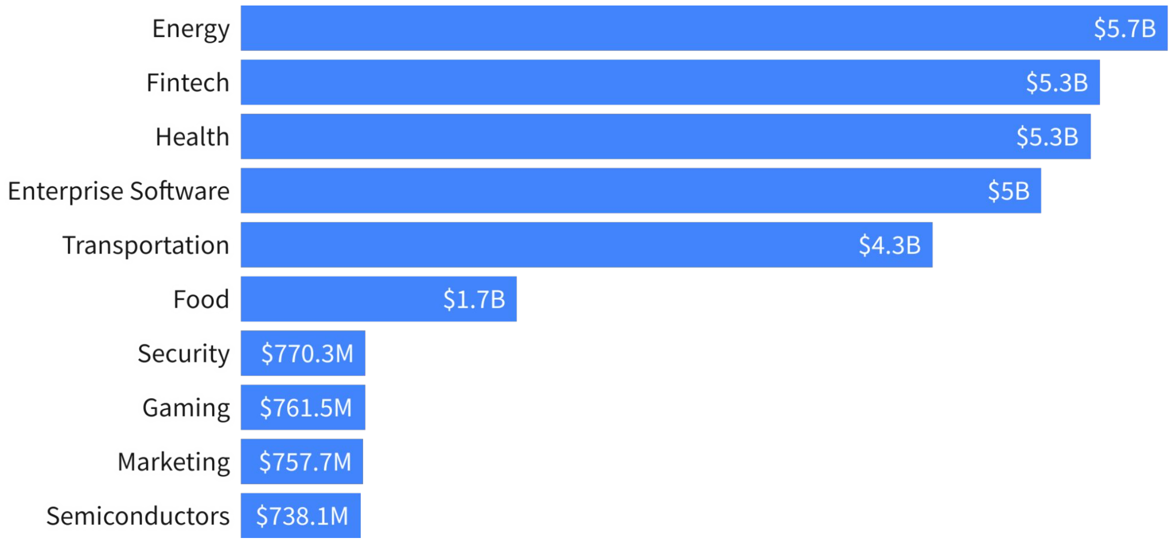



Source: Dealroom.co
Data as of Aug 2024

# In fact Energy is the most funded sector overall in Europe in 2024.

This continues a trend from 2023 when Energy was the top funded industry in Europe every quarter.

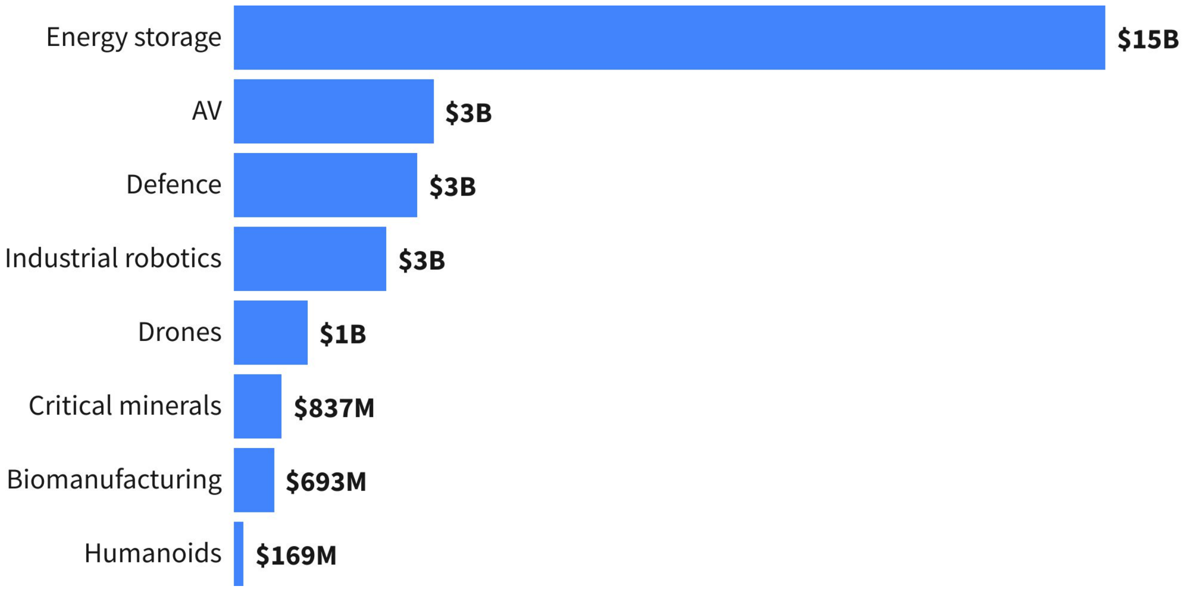
Europe's leading industries by VC investment, H1 2024




Source: Dealroom.co
Data as of Aug 2024

# Energy storage is the most funded physical tech segment in Europe.

European VC funding in selected physical segments (2016 - 2024)



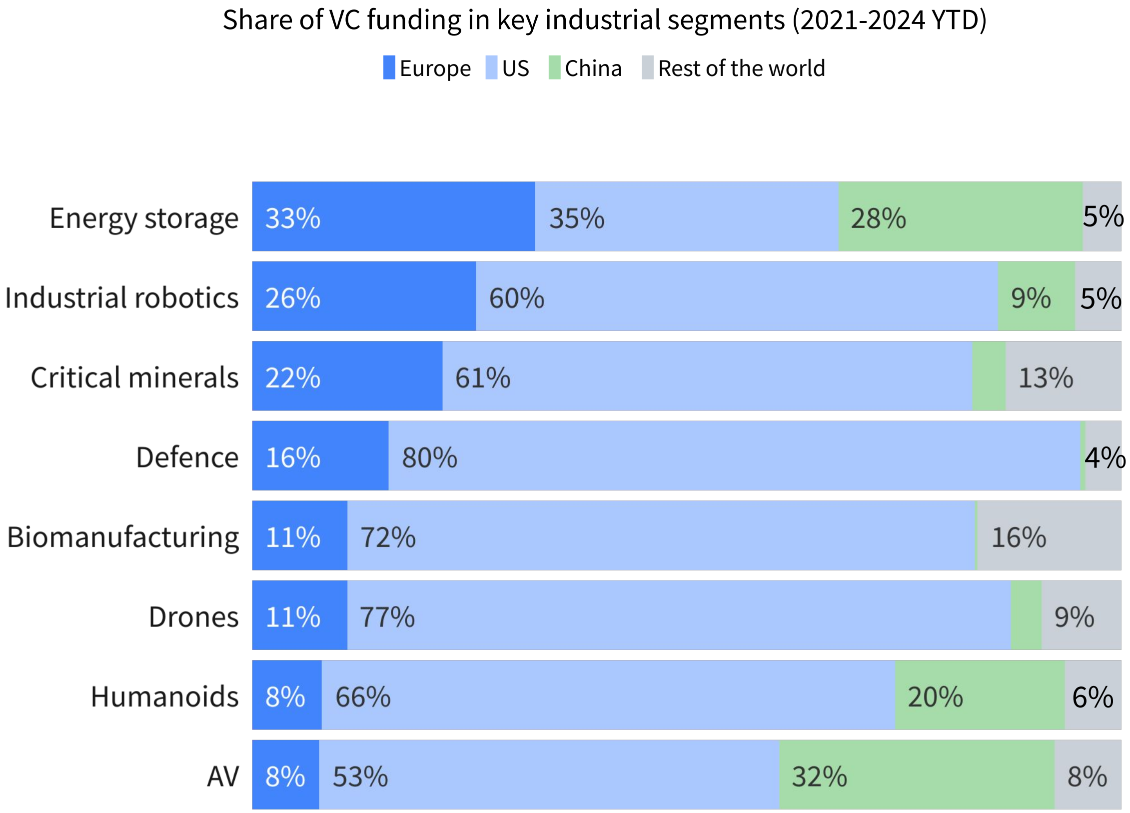

Source: Dealroom.co 2024 data as of Aug 27th 2024.



dealroom.co

# Europe is performing well in the energy storage, industrial robotics and critical minerals segments in terms of VC funding.

However, it particularly trails behind in Humanoids and Autonomous mobility.




Source: Dealroom.co
2024 data as of 27th Aug 2024

# European Industrial Robotics startups

» Explore the landscape

### Warehouse automation
Combined funding $ 802M

Zivid, QualySense, Exotec, VersaTile Autom, Zeal Robotics, SysDesign, Enabl Technolog, Dexory, ANTICIPATE GmbH, Pixel Robotics, Ubyko, Guidance Automa, Aeriu, Bottobo Robotic, Deus Robotics, Add'Verb, NODE Robotics, coboworx, Euclid Labs - 3, Startiun, Meshine Swarm T, Illumo Robotics, Flowcopter, Robotic Innovat, Noyes, ENERMOVE, Fox Robotics, SILEANE, Coalescent Mobi, Pickr.AI, Asyril, Agilox, KFactory, Synaos, GenieConnect, Robodev GmbH, Scallog, Versabox, Boa Concept, E-cobot, TruPhysics, Franka Emika, IFollow, Follow Inspirat, EffiMat Storage, Gideon, Arculus, Enway, Mobile Industri, Stanley Robotic, Balyo, Tecnimusa, Kinalisoft s.r., Milvus Robotics, Nomagic, GOGOA Mobility, Anronaut, PAL Robotics, Intelligent Rob, Effidence, Delta Drone, Fizyr, RoboCV, Smart Robotics, Verity, Cellumation, Tractonomy Robo, Flexlog, Minervasys, Magazino, EyePick, ANAVIA, RemBrain, ALTO Robotics, PHINXT Robotics

### Manufacturing and assembly
Combined funding $ 472M

Diota, RobCo – The Rob, Agile Robots, Kernova, POMO Robotics, ziknes, Motion-Robotics, BotX, Kassow Robots, Archipelago Tec, Ars Ingenii, Bitrobotics Ind, Vanguard Automa, ABAGY, Robotics Softwa, Gizelis Robotic, Inrotech, ISYBOT, Moi Composites, Prodrive Techno, ESKITEC GmbH, silana, Saeki Robotics, Cognivix, Almac - Italia, Automata, Designpro, Trussmatic, Blue Workforce, Convergence Ind, Triditive, Nimbl'Bot, Infomatika, Daedalus, Q5d, Oversonic, MX3D, Universal Robot, A3L, AIXEMTEC, Inovo Robotics, AlumatIQ, i2U, voraus robotik

### Facility inspection and maintenance
Combined funding $ 304M

NavVis, Inspekto, Invert Robotics, Acwa Robotics, ANYbotics, Ethon AI, Hummingdrone, DiagRAMS Techno, Flybotix, SubBlue Robotic, BladeBUG, Elwave, Aerones, Cin Advanced Sy, ORCA Hub, TriVision, GE Inspection R, Advanced Mechan, Distran, Aerial Power, MAB Robotics, Industrial Visi, Scopito, Sentispec, Keybotic, InnoTecUK, Gearbox Innovat, Autonomous Indu, Covision Qualit, Pipein, AIS Field

### Construction
Combined funding $ 91M

Monumental, Automated Archi, Q-Bot, Vemcon, Odico, KEWAZO, Hyperion Roboti, Robot At Work, Les Companions, Aim Robotics, hyperTunnel, Spin Robotics, Kobots, Rebartek, Rope Robotics, HAL Robotics, Northsence AB, MESH AG, Tarracor

### Agriculture
Combined funding $ 330M

Saga Robotics, Agrointelli, Stenon, Small Robot Co, ZoomAgri, Caterra, listt.io, Smart Farm Robo, Terra Robotics, Görsentam, INVERTO, Cando Drones, AutoPickr, Dücker Robotics, Antobot, Greenfield Tech, Agerpix, fruitcore robot, Ullmanna, Naust Robotics, Floating Roboti, BioBee Technolo, Fieldwork Robot, AISPRID, Osiris Agricult, Muddy Machines, Saia Agrobotics, Viridius, ENKITEK, Crover, Flox, Ekobot, Dogtooth Techno, Earth Rover, Autoagri, Agrolabs, Earth Rover, Seasony, Tibot Technolog, Robotnik, Odd.Bot, Kind Technologi, Ecorobotix, Toutiterre.Fr, RoBoa, Drone Ai Servic, Naïo Technologi, Cordon Technolo

### Recycling
Combined funding $ 49M

RECYCLEYE, Sadako Technolo, ZenRobotics, Danu Robotics, Project.BB, RFND Technologi, Picvisa, Spacedrip, REP-TEC, FPD Recycling, Antfarm

### Mining
Combined funding $ 13M

Zyfra, Oinride, HLS Hard-Line S, Gravis Robotics, Inkonova

### Logistics
Combined funding $ 258M

Cartken, Swiss-Mile, Elemental, Robolem, Innok Robotics, Swarm Logistics, ex9, Borobo, Sky-Drones, Generic Intelli, 1X Technologies, Airolit, Cleveron, Delivery Couple, THEO, EasyMile, GIM Robotics

### Restaurant tech
Combined funding $ 88M

Aitme, Karakuri, Cala, Bolk, Yanu, GoodBytz, KAIKAKU, EBar Initiative, Pizza Fleet, Qualityfry, Cook-e, Ceratech, Pazzi, Glanola, Ztove, Natural Machine, MIXARTISTA, Retail Robotics

### Defence
Combined funding $ 26M

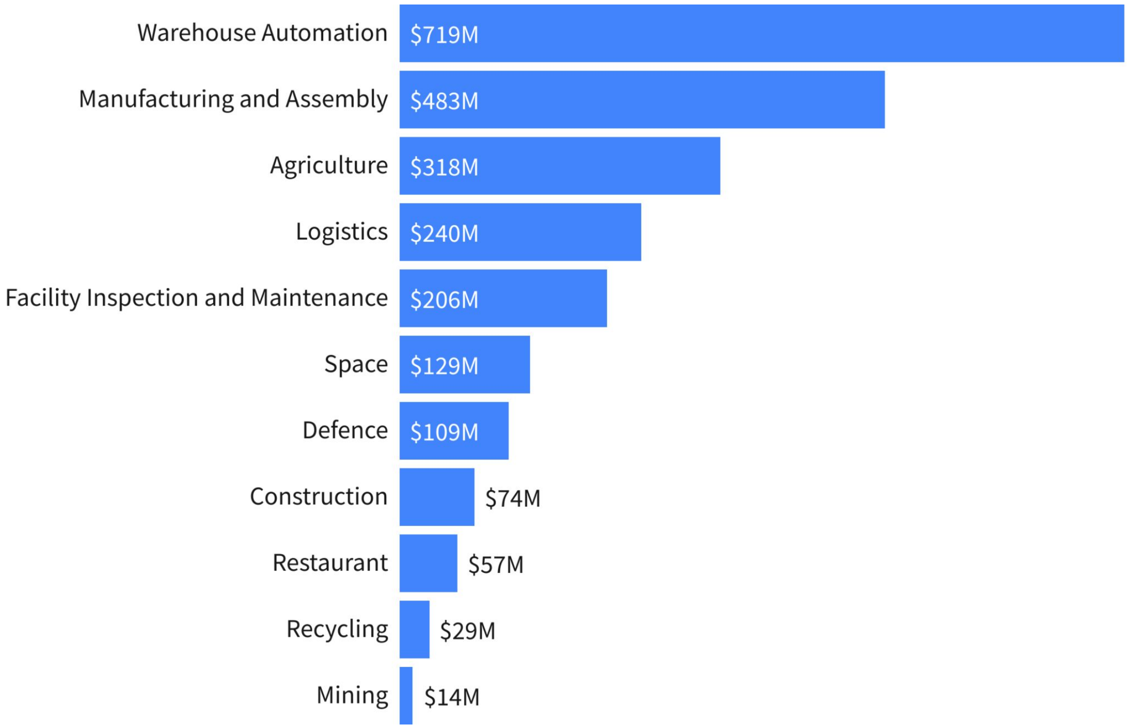
Milrem Robotics, Delfox, ARX Robotics, Shark Robotics, Stanhope AI, BlinkTroll Robo

### Space
Combined funding $ 17M

Space Mobility, KINETIK Space, XSun, Kurs, AHYRES, SpaceBIT, Ntention

# Looking at European industrial robotics, warehouse automation is the most funded segment, followed by manufacturing and assembly robots.

VC funding per European industrial robotics segments (2020-2024 YTD)




Source: Entropy Capital, Dealroom.co
Data as of Sep 12th 2024.

# European Industrial Logistics startups

» Explore the landscape

### Drones
Combined funding $ 200M

Uavia, Animal Dy…, Manna Dr…, Connect …, Diaven, Lorenz Te…, Skyports, RigiTech

### Autonomous Vehicles
Combined funding $ 1.9B

Academy o…, Eliport, TeleRetail, Twinsw…, Oxa, EasyMile, Swarm L…, Sensible 4, Immense S…, Einride, Wayve, Fernride

### Warehouse Automation
Combined funding $ 615M

Gideon, Exotec, Magazino, Balyo, Enway, Fizyr, Smart R…, Intellige…, Stanley …, RoboCV, Mobile I…, Tecnimusa, Laevo, Transolt, Unchained…, GOGOA…

### Tracking & Visibility
Combined funding $ 411M

Wakeo, BIOTRAQ, Onalytics, CargoSt…, Live Trac…, Shippeo, Virtenio …, Traxens, Arviem, Hispatec…, FastTrack …, Controlant, Dockflow, Modum, Ambrosus, Everysens, Undagrid, Packwise, Tag Sens…, BOX ID S…, Most, Synfioo …, Nexxiot, ScanTrust, Minespider, Logmore, Tilkal, Viloc, HiveTrac…

### Supply Chain Management
Combined funding $ 1.3B

Zig-Zag, Cargofive, TradeLink, Maindeck, Akanea, Izifret, Shipamax, SupplyS…, VEKIA, Lexi Sol…, Portchain, Shipfix, Routal, Winddle…, Riskmet…, Nova Tra…, Catkin, Obuu, ItsMyCargo, Sendify, Freightly, Tiramizoo, CargoSnap

### Smart Pallets & Containers
Combined funding $ 211M

Aeler, Navlandis, SkyCell, Ahrma P…, LNGTainer, PALLETE…

### On-demand storage
Combined funding $ 41M

FLOX, Stockbo…, DepotCity, Store-An…, SpaceFill, Waredock, Wareho…

### Shipping Marketplace
Combined funding $ 76M

HUBBIG, Shipwise, Upela, Ontruck, CheapCar…, TRANSPORT…, Quicargo, Dliver, STYX, Transeop, BIZCARGO, Pamyra, ZeroNorth

### Logistics Procurement
Combined funding $ 277M

Cargo One, XChange, PackLink

### Freight Forwarding
Combined funding $ 3.7B

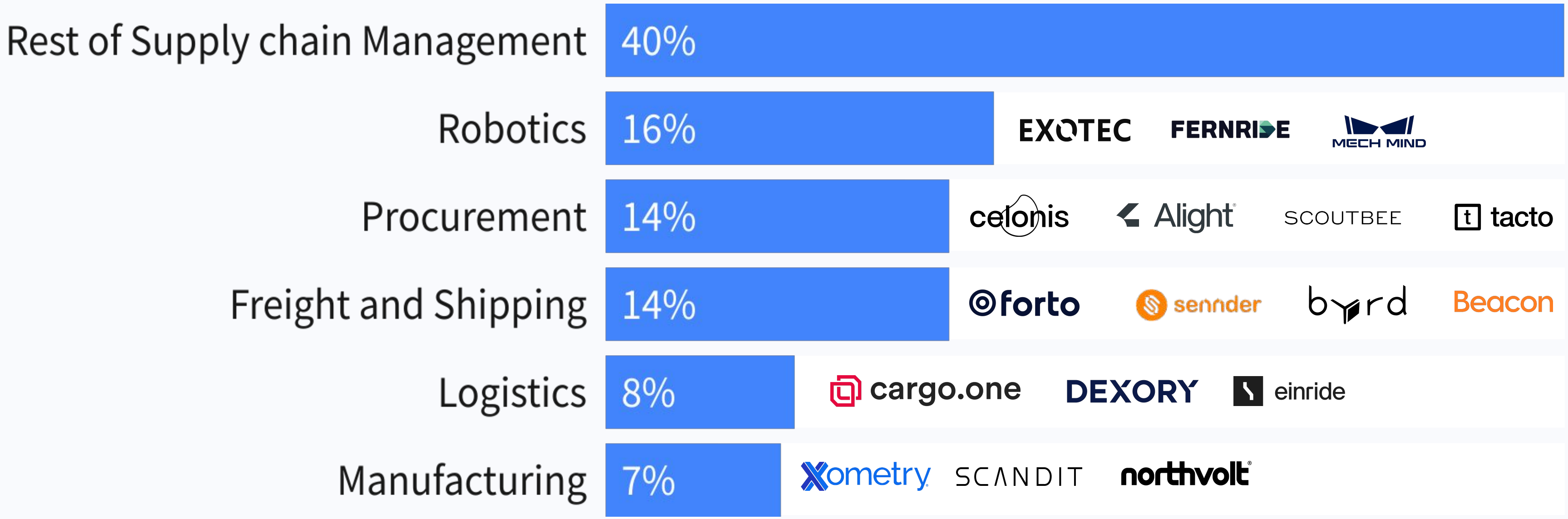
Trusk, Freightol, Packbud, Truckin, Cargonexx, Saloodo, Mub Cargo, Logisee, Sennder, Chronot…, Ovrsea, Carrypic…, Tellus Lo…, Trucksters, Kontainers, Forto, Fretlink, Everoad, Beacon, Zencargo, InstaFrei…, Shypple, Flexport, Deliver.ru, Flying W…

### Ecommerce Logistics
Combined funding $ 1.2B

Cubyn, Seven S…, Sorted G…, Lineten, Sendcloud, ParcelBr…, Airmee, Byrd, Weengs, Omnipack, Codept, Paack, URBANTZ, Huboo, Webshipp…, ParcelLab, DroppX, HUUB, ShippyPro, Bigblue, Kubbo, Imagine…, Annanow G…, Wing

# Since 2020, supply chain startups have raised $14B in VC funding, with a significant portion of which allocated to robotics.

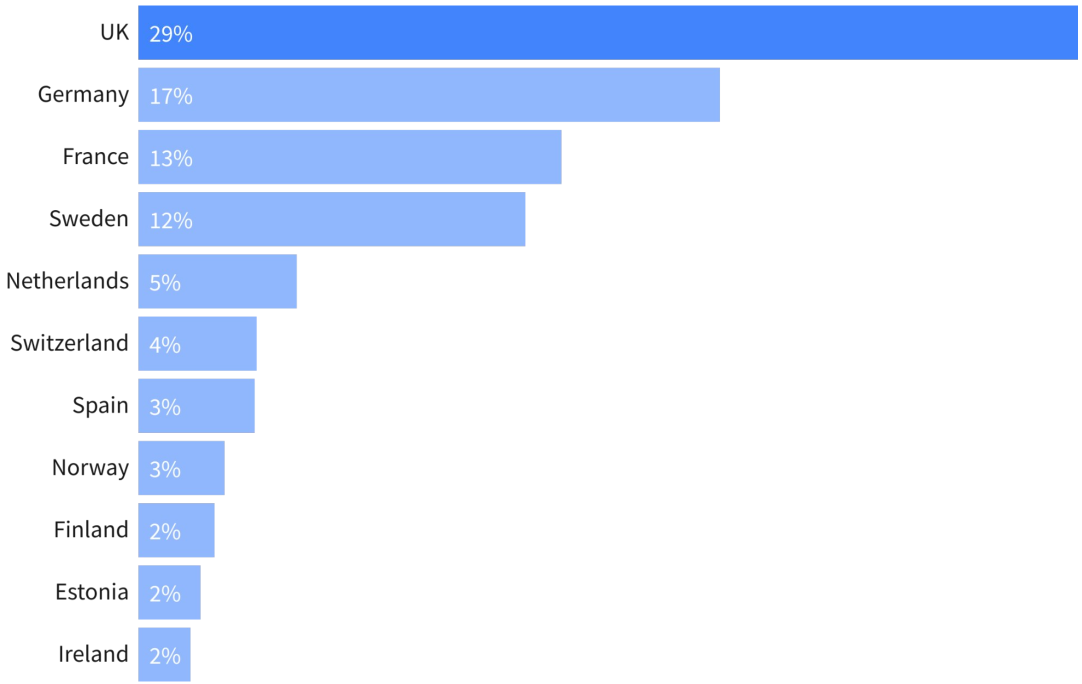
Supply chain management funding by segment (2020-2024 YTD)




Source: Entropy Capital, Dealroom.co
Data as of Aug 29th 2024

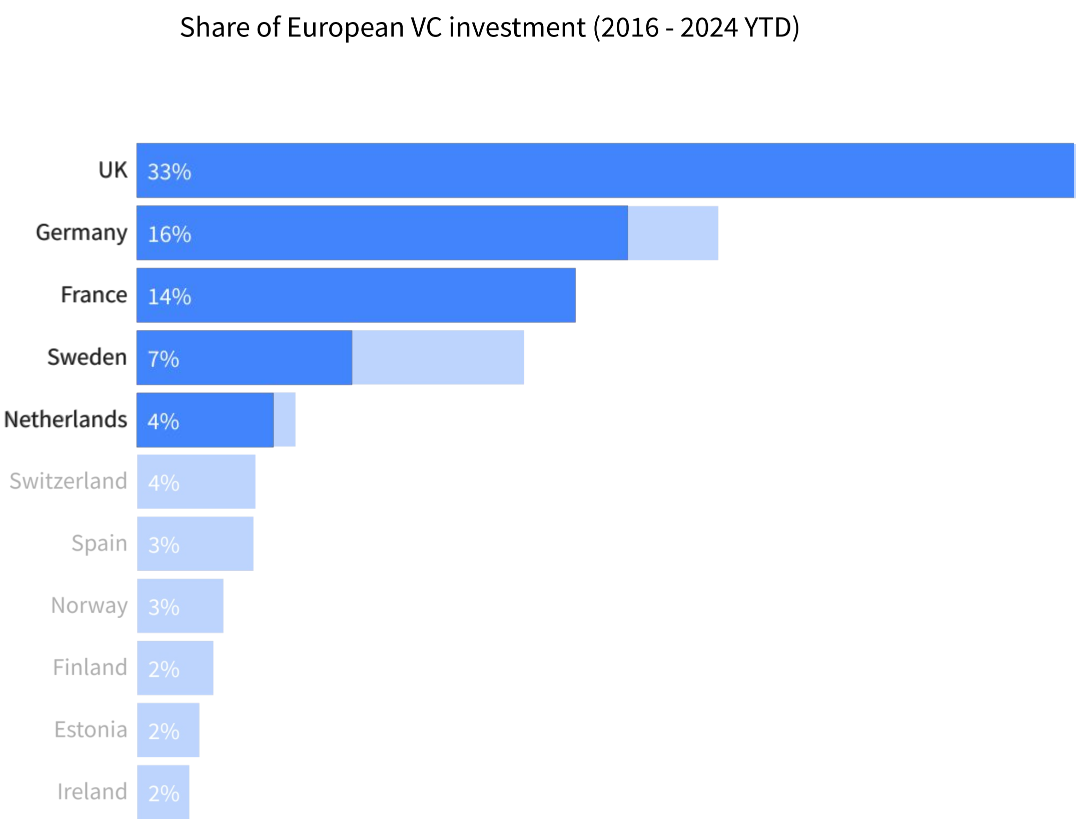
# The UK has raised 29% of European physical industry funding.

Share of European VC investment into physical industries (2016 - 2024 YTD)




Source: Dealroom.co
Data as of Sep 2024

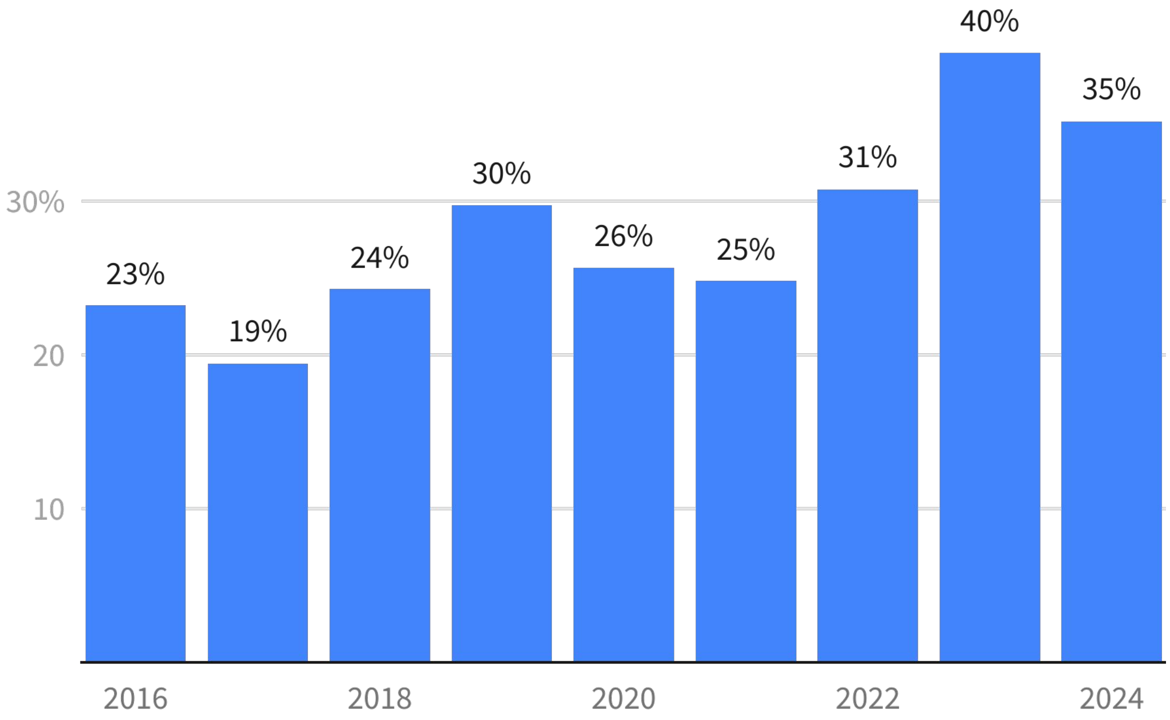
# The UK is less dominant in industrial tech than the rest of tech.




Source: Dealroom.co
Data as of Sep 2024

# European tech is undergoing a gradual industrial pivot.

% of European VC funding going to "physical industries"




Source: Dealroom.co
Data as of Aug 2024

# Selected European Physical startups

» Explore the landscape

### Energy
Combined funding $ 10.3B

Northvolt, Verkor, cylib, BASQUEVOLT, Energy Dome, Instagrid, Morrow, Nyobolt, Ingrid Capa…, Mecaware, Kraftblock, Enerpoly, Nexeon, NHOA, VoltStorage, Elestor, GridBeyond, Arkion, Synaptec

### Supply chain management
Combined funding $ 1.4B

Prewave, Scoutbee, osapiens, Tacto, Rackbeat, Controlant, Relex Soluti…, Xeneta, Sievo, Katana, SpaceFill, Ignite, Archlet, Shipsta, INTRANAV

### Manufacturing
Combined funding $ 2.7B

Caracol, Workerbase, 3Deus Dyna…, gemineers, XTPL, ICOMAT, Neuron Sou…, ATLANT 3D, Aibuild, Thingtrax, Braincube, OxMet Tech…, CNCTeile24, Metaloop, H2 Green St…

### Robotics and automation
Combined funding $ 1.1B

1X Technolo…, ANYbotics, Agile Robots, Exotec, Magazino, RobCo – Th…, Invert Robo…, Verity, Blue Danub…, Ethon AI, Mobile Indu…, Saeki Robot…, Nomagic, SILEANE, silana, Covision Qu…, Oversonic

### Logistics
Combined funding $ 3.3B

Flexport, ZeroNorth, Flying Whales, Fernride, Aeler, Einride, Ontruck, Sennder, VONZU Tech, Chronotruck, Kardinal

### Aerospace and defence
Combined funding $ 2.3B

Isar Aerospace, The Explora…, Kuva Space, QuadSAT, ReOrbit, Aldoria, Helsing, Quantum-S…, TEKEVER, Unseenlabs, Faculty, Scaleout Sy…, Rocket Fact…, E-Space, Reflex Aeros…, Destinus, Latitude, PLD Space, ARX Robotics, STARK, Buntar Aero…, Osavul, Fossa Systems, EnduroSat, Orbex Space, Open Cosmos, ALL.SPACE, CAILabs

### Critical minerals
Combined funding $ 151M

Cornish Lith…, MagREEsou…, Adionics

### Advanced materials
Combined funding $ 873M

Nanogence, Avantium, Again, OCSiAl Group, Fairmat, traceless m…, Material Evo…, Paragraf, BComp, Canatu, CNCTeile 24

### Architecture, engineering and construction (AEC)
Combined funding $ 59M

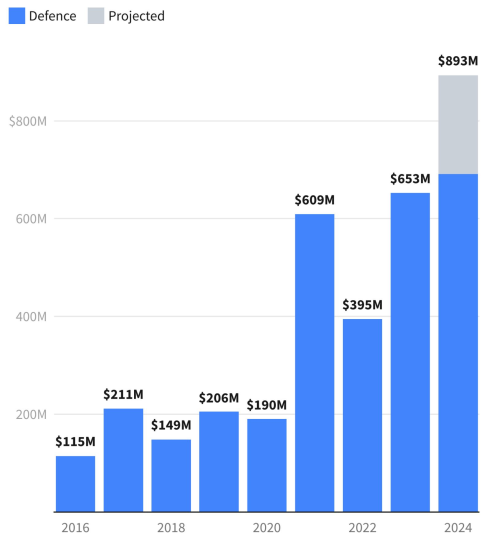
Monumental, Automated …, KEWAZO, Hyperion R…, hyperTunnel, BeyondMath

### Design Software
Combined funding $ 127M

Neural Conc…, Continuum …, Trinckle, SkAD Labs, Hyperganic, Celus, ToffeeX, PhysicsX

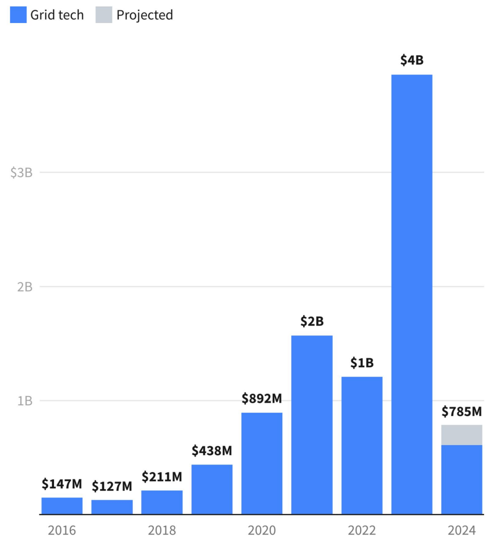
# European **defence tech**

VC investment ('24 projected)



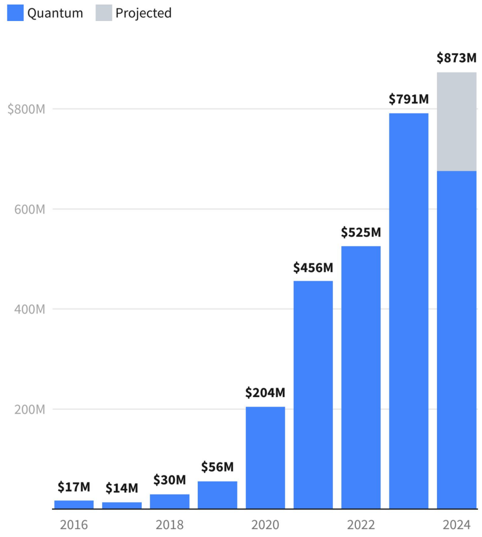

# European **grid tech**

VC investment ('24 projected)




# European **quantum tech**

VC investment ('24 projected)






Source: Entropy Capital, Dealroom.co
Data as of Sep 16th 2024.

# Global quantum computing startups

» Explore the landscape

## Quantum computers & processors

Combined funding $ 5.1B

- Oxford Quantum
- IQM
- Alice&Bob
- ORCA Computing
- eleQtron
- QuiX Quantum
- Weling
- Quantware
- Kipu Quantum
- Equal1 Laborato
- C12
- Oxford Ionics
- Universal Quant
- AegiQ
- Qilimanjaro Qua
- Quantum Motion
- Pasqal
- Delft Circuits
- Sparrow Quantum
- Qdevil
- Alpine Quantum
- NextGenQ
- Quantum Brillia
- planqc
- QuEra Computing
- Rigetti Computi
- PsiQuantum
- Xanadu
- Infleqtion
- Atom Computing
- SeeQC
- Quantum Circuit
- IonQ
- Origin quantum
- 1Qbit
- Nord Quantique
- Quantum Source
- SpinQ
- AuroraQ
- Quandela
- Kelvin Nanotech
- LightSolver
- Orange Quantum
- Aquabits
- Arque
- Beijing Bose Qu
- Diraq
- Quantinuum
- Aquark Technolo
- Qblox
- SemiQon
- QC Design
- Photonic
- Quobly
- LakeDiamond SA
- XeedQ
- Quantum Silicon
- Qphoton
- Superchips
- Nanofiber Quant
- Silicon Quantum
- EeroQ
- Bleximo
- III-V Epi
- Q-Block Computi
- Turing Quantum
- Silent Waves
- Analog Quantum
- QC82
- Huayi Boao Quan
- Anyon Systems
- Aurora Quantum
- CreoTECH
- Sivers Photonic
- Bra-Ket Science
- Wideblue

## Quantum computing software

Combined funding $ 1.6B

- Multiverse Comp
- Ketita labs
- QuantrolOx
- Quanscient
- Phasecraft
- Cambridge Quant
- Nordic Quantum
- QuSoft
- JoS Quantum
- Qu & Co
- Riverlane
- Quantastica
- Beit
- QMWare
- ParityQC
- Quantagonia
- Fermioniq
- Kvantify
- SandboxAQ
- Classiq
- D-Wave Systems
- Quantum Machine
- Zapata
- QC Ware
- QxBranch
- Horizon Quantum
- Qindom
- Dark Star Quant
- Strangeworks
- MDR
- Entropica Labs
- QbitLogic
- Agnostiq
- BosonQ
- Super.tech
- Quantum Benchma
- SoftwareQ
- Xofia
- Quantum Flytrap
- Quanvia
- PlanQK
- Ingenii
- M-labs
- A-Star Quantum
- ColibrITD
- Quantum Computi
- Artificial Brai
- Qtft
- Infinityq
- Qunova Computin
- Solid State AI
- Foqus
- Adaptive Financ
- Qpi Technology
- Next Generation
- Haiqu
- RealQ
- Qudot
- Qrithm
- Labber quantum
- Q-Lion
- Turing Quantum
- Elyah
- Tokyo Quantum C
- Quantfi
- Boxcat
- Quantum Mads
- Black Brane Sys
- Signaloid
- Quantum Insight
- QuantumSense
- Semicyber
- Qedma

## Quantum cryptography

Combined funding $ 344M

- Terra Quantum
- KETS Quantum Se
- Arqit
- QuSide
- Crypto Quantiqu
- Quantum Base
- iQrypto
- Quantum Dice
- ThinkQuantum
- Nu Quantum
- Infiniquant
- CryptoNext Secu
- ID Quantique
- Keequant
- Crypta Labs
- LuxQuanta
- Quantum Xchange
- QuintessenceLab

## Post-quantum era encryption

Combined funding $ 107M

- Crypto4a
- Chelpis
- QuantiCor Secur
- BLAKFX
- Post-Quantum
- PQShield
- CyberHive
- ISARA Corporati
- Cyph
- Resquant
- TACEO

## Quantum computing for biotech and chemical applications

Combined funding $ 158M

- Algorithmiq
- Qubit Pharmaceu
- Rahko
- HQS Quantum Sim
- Quantistry
- Hafnium Labs
- Pharmacelera
- Kuano
- Proteinqure
- Materials Nexus
- Quantum Simulat
- Aqemia
- Molecular Quant
- ApexQubit

## Quantum communication

Combined funding $ 50M

- QphoX
- Qunnect
- BraneCell Syste
- Aliro Technolog
- MemQ
- Quantum Bridge
- Craft Prospect
- Qubitekk
- Synergy Quantum
- Quantum Optics
- Icarus Quantum
- Ki3 Photonics
- Entangled Netwo
- LQUOM
- SQE
- Q*Bird
- Qubit Reset

## Quantum sensing

Combined funding $ 211M

- Qnami
- QustomDot
- Muquans
- Q.ant
- QuantumDiamonds
- Q-CTRL
- Vector Atomic
- QDTI
- Quantum Solutio
- Adamant Quanta
- QLM Technology
- MIRO Analytical
- NVision Imaging
- QDI-Systems
- NIQS Technology
- Guardion
- Nomad Atomics
- QuantaMap
- SBQuantum
- High Q Technolo
- Quantum Diamond
- Atomionics
- InSpek
- Hoptroff

## Photon detection & counting

Combined funding $ 13M

- Quantum Detecto
- Single Quantum
- Photon Force
- Micro Photon De
- Pixel Photonics
- Universal Quant
- ElFys

## Photonic integrated circuits & photonics IP

Combined funding $ 438M

- Salience Labs
- Ligentec
- EFFECT Photonic
- SMART Photonics
- ActLight
- Scintil Photoni
- Photonpath
- LightOn
- Optoscribe
- Ipronics
- Lumai
- Akhetonics
- Black Semicondu
- CamGraPhIC
- Polariton Techn
- Optalysys
- Enlightra

## Quantum cascade lasers, laser tech for sensing & LiDAR

Combined funding $ 25M

- Chromacity
- Integrated Opti
- Pilot Photonics
- Lytid
- Eblana
- QuantaRed Techn
- Sicoya
- Pranalytica

## Next gen laser & photonics

Combined funding $ 82M

- M Squared Laser
- VEXLUM
- Skylark Lasers
- Spark Lasers
- Living Optics
- Vector Photonic
- Fusion Bionic G

## Refrigeration for quantum

Combined funding $ 29M

- Zero Point Cryo
- Kiutra
- Maybell Quantum
- Bluefors
- Leiden Cryogeni

# European Defence startups

» Explore the landscape

### Command, Control, Communications, Computers, Intelligence, Surveillance and Reconnaissance - C4ISR
Combined funding $ 275M

SensusQ, Adarga, Inex Microtechn, Revobeam, AV Optics, SECQAI, QuadSAT, Siren, Living Optics, Disruptive Indu, Matrix.org, Labrys, Element, Silicon MicroGr, Scalinx, Himera, Watch Bird, KNL Networks, Rcam, RFence, Focal Point Pos, Evitado Technol, Ajax Systems

### Weapons/Defence Systems
Combined funding $ 28M

Milrem Robotics, MyDefence, Aktyvus Photoni, Gwagenn, MirSense, SINTERMAT, Anybody Technol, Zvook, Ytsab Defence, Operational Sol

### UAVs
Combined funding $ 244M

Lambda Automata, Quantum-Systems, ARX Robotics, Elistair, Alpine Eagle, Arktis Radiatio, KrattWorks, Unmanned Life, Zepher Flight L, Dronetag, TEKEVER, Shark Robotics, STARK, Baykar Technologie, Novadem, Unmanned Defens, Threod Systems, Angryeyes, SKAITECH, Aurea Avionics, Greenjets, RSI Europe, WARGdrones, Delair, Robot Aviation, Velos Rotors, Origin Robotics, Tidav, DroneUA, Avalor AI, HIGHCAT, Point Zenith, Orqa FPV, Skya, Sky-Watch, Nordic Unmanned, Monopulse, Broswarm, Xplora Srl, Roboneers

### Naval and Maritime Technologies
Combined funding $ 34M

Rovco, dotOcean, Skarv Technolog, LOBSTER Robotic, Elwave, Sotiria Technol, Water Linked, SEABER, Hefring Marine, Havguard, Maritime Roboti

### AI x defence
Combined funding $ 941M

Helsing, Stanhope AI, Comand AI, LatticeFlow, Grayscale AI, Preligens, Hala Systems, Blackshark.ai, Arondite, Invisum AI, Labelfuse, Advai, Materials Nexus, Neurobus, GScan, Mission Decisio, Scaleout System, Unbound Autonom, TYTAN Technolog, Buntar Aerospac, GIGA Venture, Ask for the moo, Defsecintel, Sky Engine AI, Archangel Imagi, Faculty

### Training and Simulation
Combined funding $ 15M

Hologate, Skyral, Levato AS, 4C Strategies, Sense Glove, VRAI, PSS by Logics 7, BlinkTroll Robo, IMPETUS AfeaS, Drill, EODynamics, MXR Tactics

### Space and Satellite
Combined funding $ 1.2B

ALL.SPACE, ICEYE, Unseenlabs, The Exploration, Reflex Aerospac, Isar Aerospace, CAILabs, Spacelis, Astrolight, Space Forge, Dark, Planetek Hellas, Tyvak Internati, Fossa Systems, Krono-Safe, Satellite Vu, GomSpace, Satcube, Remos, Space Apps

### Supersonic/ hypersonic planes and propulsion systems
Combined funding $ 72M

Destinus, Aeralis, Hypersonica, Axter, Magdrive

### Quantum computing, cryptography and sensing
Combined funding $ 416M

Terra Quantum, Quandela, Nu Quantum, KETS Quantum Se, QuSide, Quantum Dice, CryptoNext Secu, Qubitrium, levelQuantum, G2-Zero, Ephos, AegiQ, Q*Bird, IQM, eleQtron, Qubalt

### Cybersecurity and critical infrastructure
Combined funding $ 179M

Countercraft, Goldilock, Anzen, 42Crunch, EclecticIQ, ByzGen, Lab 1, Arqit, 3IPK, Januus, Runecast, CyNation, DataFlowX, DeNexus, Osavul, Modirum

### Strategic semiconductors
Combined funding $ 81M

Kalray, Aquark Technolo, Swegan, Arceon B.V., Black Semicondu, Nordic Air Defe

### Robotics
Combined funding $ 117M

ANYbotics, RobCo – The Rob, GIM Robotics, Vimotek AB

### Energy (e.g. energy storage)
Combined funding $ 62M

LeydenJar Techn, Zelestium Techn, Ore Energy, McGuire Aero Pr, Advanced Materi, Kitepower, Ionate, Wpe Research &, Icewind, GaltTec, Nordic Batterie, Molyon

### Biodefence
Combined funding $ 26M

Ionlace, Neuromorphica, Aquila Bioscien, delox, Fabentech

### Advanced materials and manufacturing
Combined funding $ 37M

ICOMAT, DECPT, ATLANT 3D

### Wearables for Military
Combined funding $ 5.2M

Standard Fighti, Aristeia, Esper Bionics

### Advanced Sensing Technology
Combined funding $ 11M

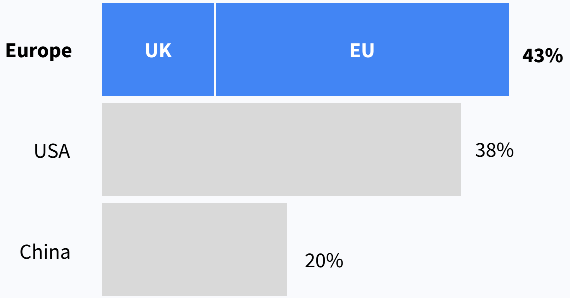
Sensrad, ElFys

### Service Providers
Combined funding $ 15M

DigitalPlatform

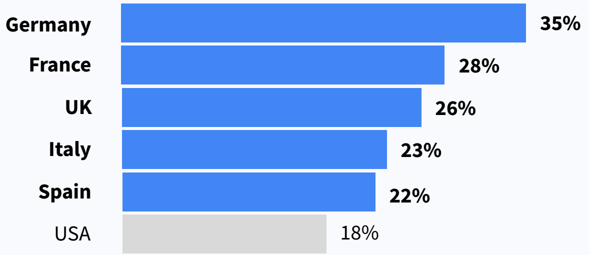
# Europe has the great technical talent and research leadership required to play a key role in the future of global industrial tech.

## High share of highly-cited research publications*




## European students are more into science*

(portion of graduates in Science, Technology, Engineering, and Mathematics)




## Europe excels in Computer Science ranking (THE 2022)

| | |
|---|---|
| **1. Oxford** | **11. Imperial College London** |
| 2. Stanford | 12. UCLA |
| 3. MIT | 13. Tsinghua (Beijing) |
| **4. ETH Zurich** | 14. Caltech |
| **4. Cambridge** | **15. TU Munich** |
| 6. CMU | 16. Singapore Nanyang |
| 7. Harvard | 17. University of Washington |
| 8. Berkeley | 18. Cornell |
| 8. National University of Singapore | 19. Peking university |
| 10. Princeton | **20. École polytechnique fédérale de Lausanne** |

## … and does well also in Engineering ranking (THE 2022)

| | |
|---|---|
| 1. Harvard | 11. University of California |
| 2. Stanford | 12. Peking university |
| 3. Berkeley | **13. Imperial College London** |
| 4. MIT | 14. Georgia Tech |
| **5. Cambridge** | 15. Singapore Nanyang |
| **6. Oxford** | 16. Yale |
| 7. Princeton | 17. Tsinghua (Beijing) |
| 8. Caltech | 18. Carnegie Mellon University |
| **9. ETH Zurich** | **19. École polytechnique fédérale de Lausanne** |
| 10. National University of Singapore | 20. University of Michigan-Ann Arbor |

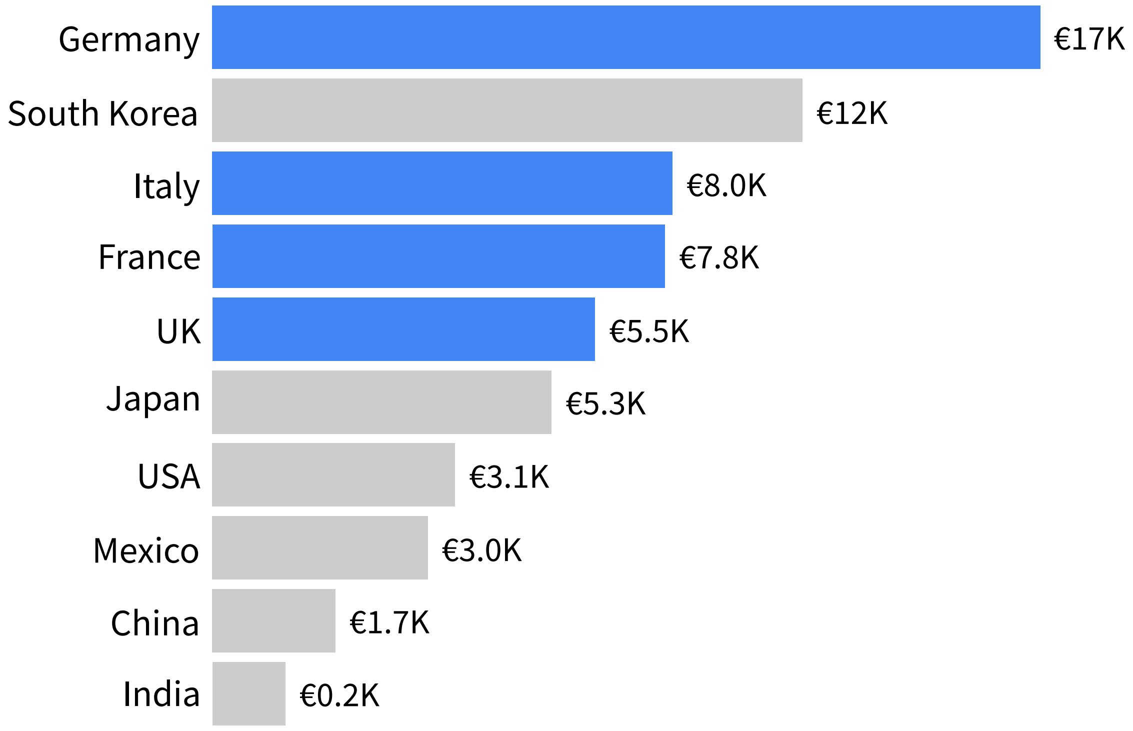


Source: * * Benedict Evans / Mosaic Ventures Dec 2020. Data originally from BEIS/SCOPUS/UNESCO.

# Germany, Italy and France are among the biggest exporters per capita globally (driven by automotive and machinery).

Global rank of manufactured exports per capita

*10 biggest manufacturing countries - UN and World Bank data*



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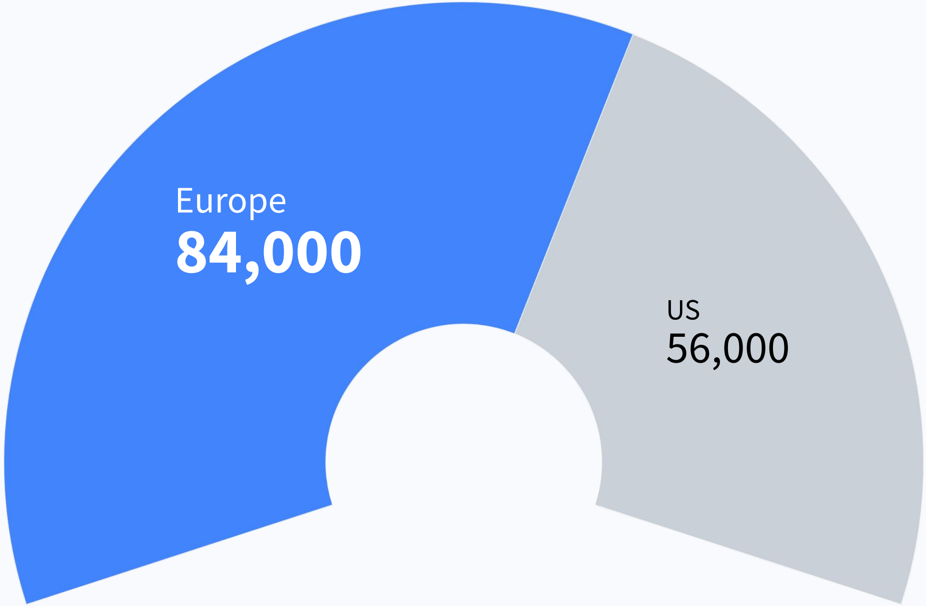
# Europe excels in manufacturing and automation, surpassing the US by number of new robots installed.

New robots installed (2023), IFR






Source: IFR, Entropy Capital, Dealroom.co

# European corporate expenditure dwarfs VC.

**50 million**
employees

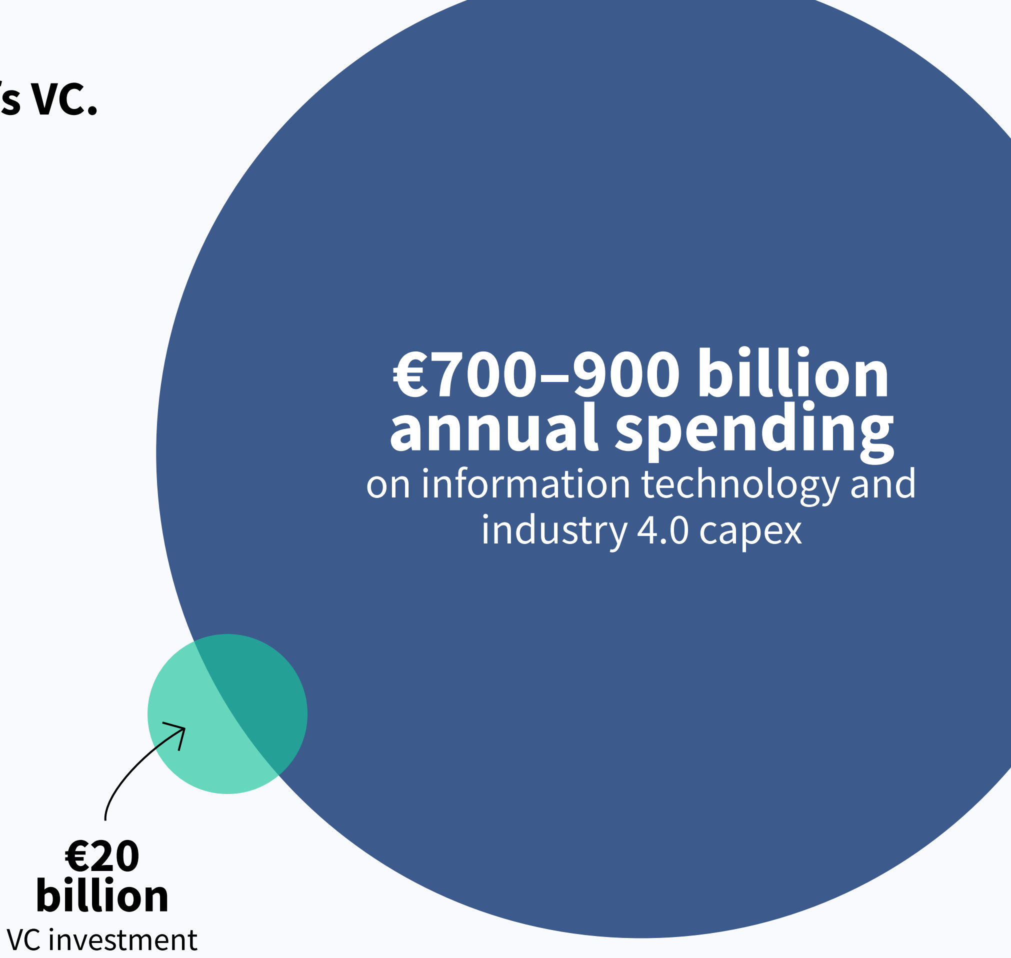
**€13 trillion** turnover

(of which ±25% Value Added* to European economy as GDP)

**Spending 2.5% to 3% on Information Technology**

€ 300–400 billion per year

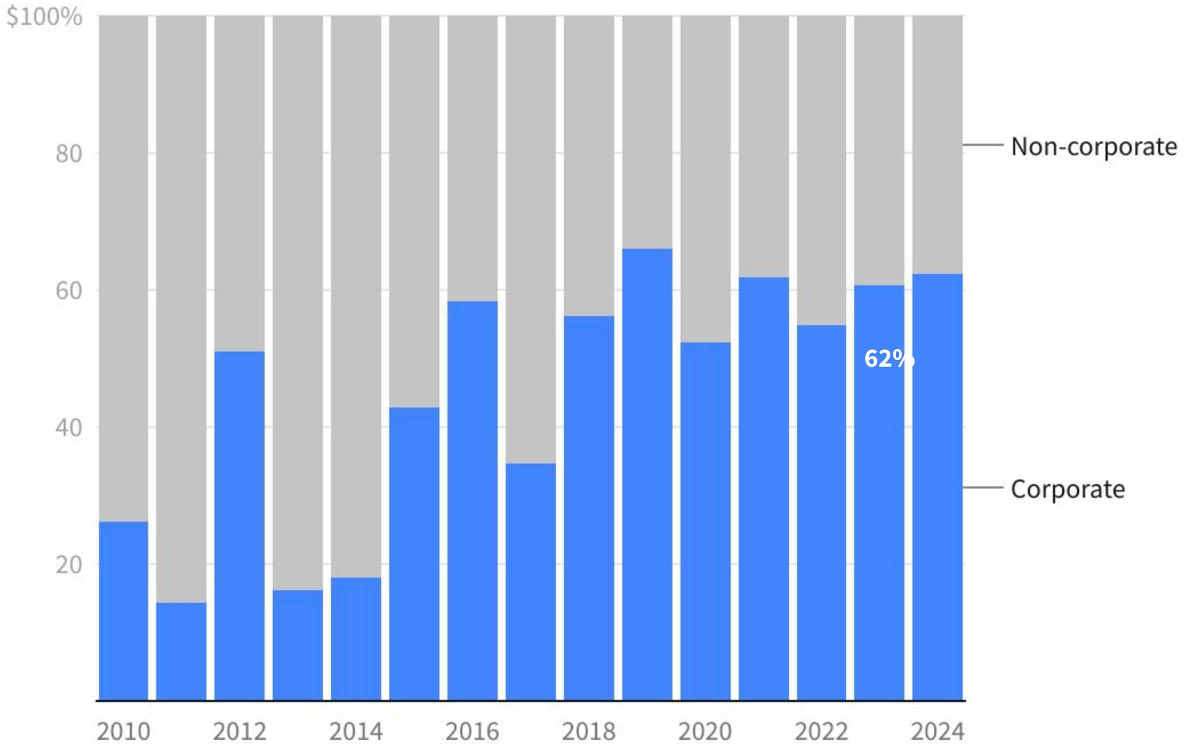
Another 3 to 4% investment into Industry 4.0 solutions
= € 400-500 billion per year




Source: Dealroom.co
* Value added is turnover minus cost of goods sold.
** Logistics are normally included in services, not industry, but it's closely related and highly relevant, therefore included here.

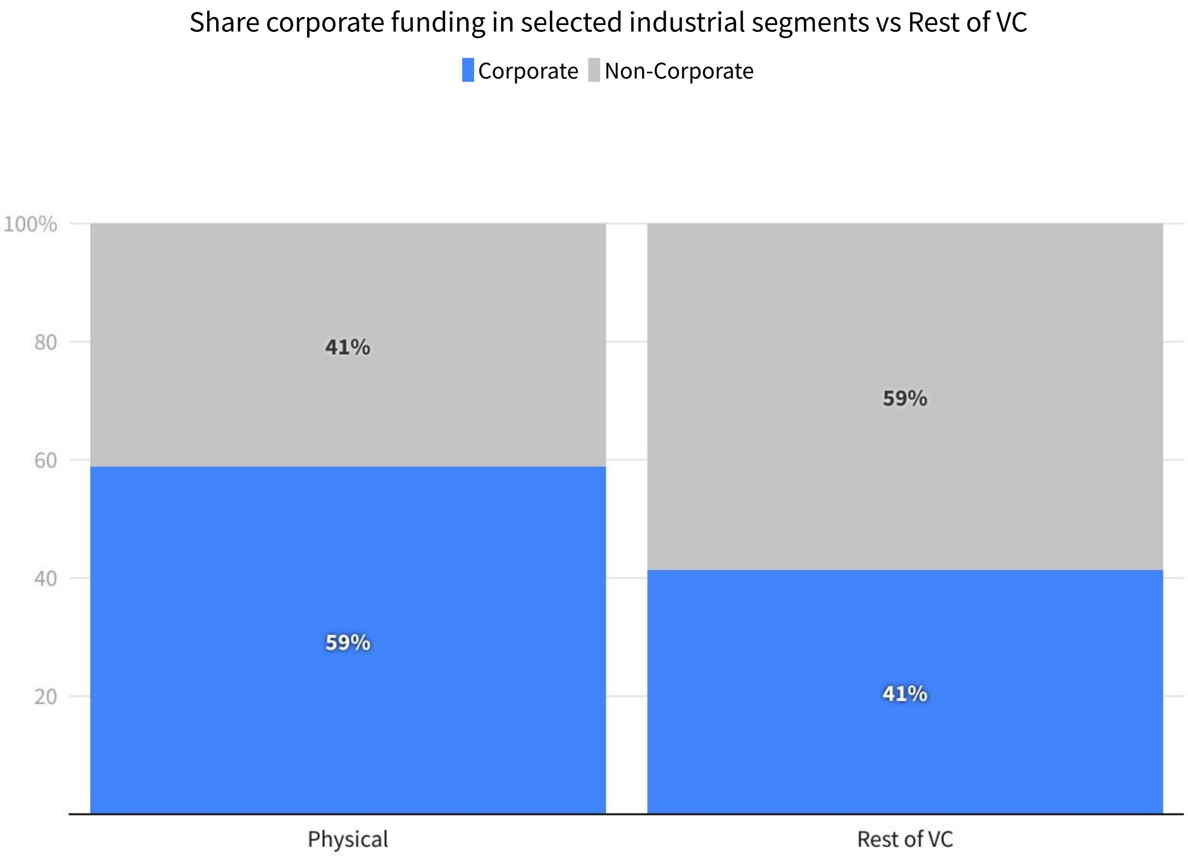
# Corporates involvement in European industrial tech VC is on the rise.

Corporate funding in selected industrial segments, Europe




Source: Dealroom.co

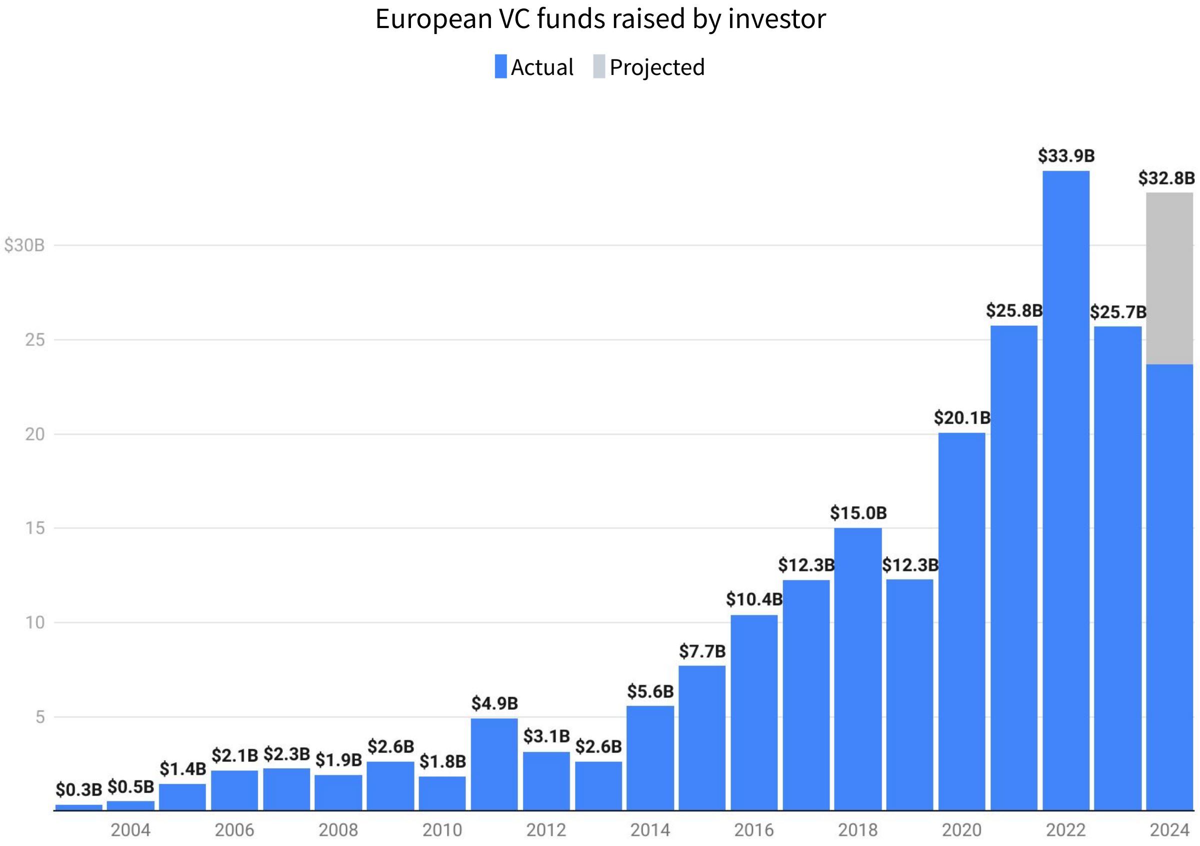
# Corporates are taking much more venture interest in Industrial tech.




Source: Dealroom.co

Record amounts of dry powder reported.

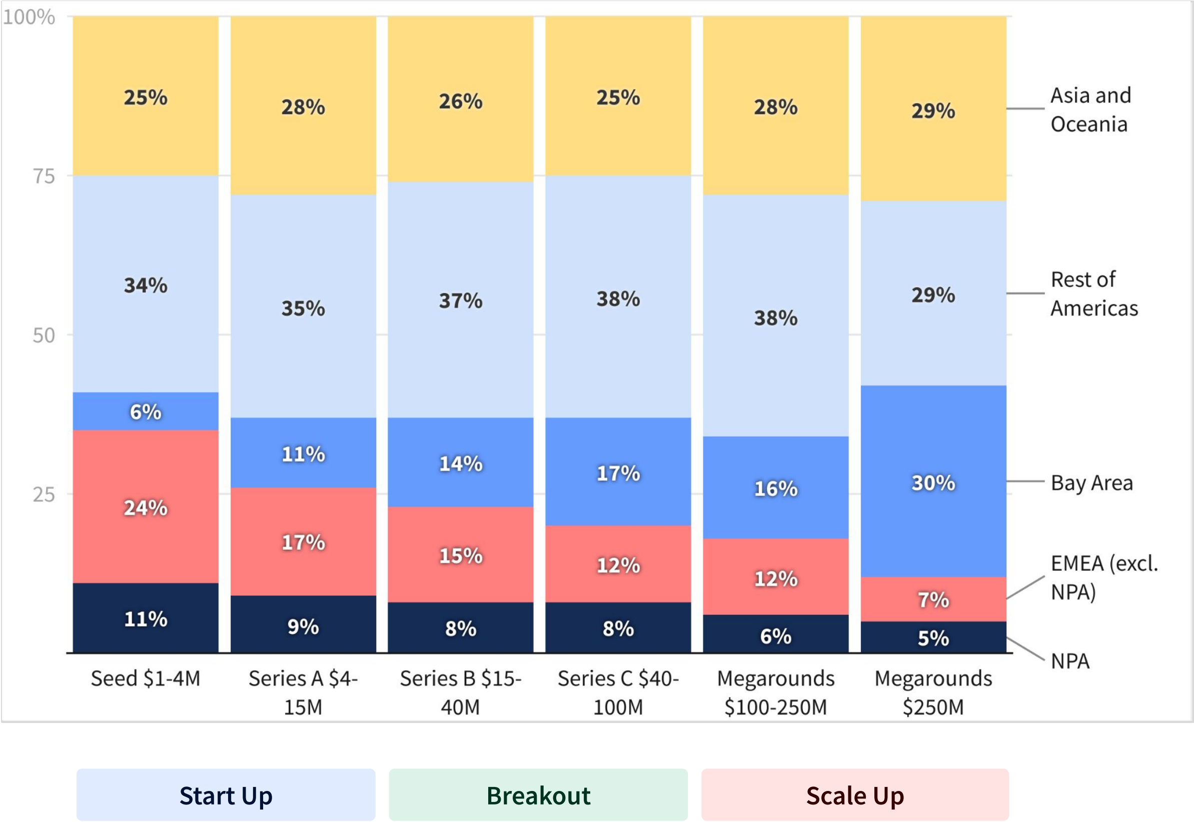
But the amount of European funds is still lags the US in terms of ecosystem size.




Source: Dealroom.co

# Despite strength at Seed & Series A, EMEA is very underinvested at later stages relative to other geographies.

Venture capital by destination and by stage (2020-2023)



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# A few words on our methodology.

### What is a startup?

Companies designed to grow fast. Generally, such companies are VC-investable businesses. Sometimes they can become very big (e.g. $1B+ valuation). When startups are successful, they develop into scaleups (>50 people), grownups (>500 people) and result in big companies. Only companies founded since 1990 are included in this report.

What is a startup?

### What is a Unicorn?

Unicorns are (former) startups that reached $1B valuation or exit at one point in time.

What is a Unicorn?

### Physical industries and industrial tech

Physical industries includes both hardware and/or software focused on physical sectors. So it includes both batteries for energy storage, but also software for grid management for instance.
Industrial Tech is in Dealroom implemented more as solutions for industrial sectors that are more transversal such as industrial procurement, industrial logistics, etc. It does not include software to manage grids or battery energy storage.

### Geographic methodology

Startups are assigned to the location of their current HQ. In case a startup moves its HQ location the change is applied in Dealroom, while the first HQ is regarded as founding location. The location of most of the employees or the founder nationality are not taken into account.

### Underlying data

Dealroom's proprietary database and software aggregate data from multiple sources: harvesting public information, user-submitted data verified by Dealroom, data engineering. Data is verified and curated with an extensive manual process.
The data on which this report builds is available via app.dealroom.co. For more info please visit dealroom.co or contact support@dealroom.co.

### Venture capital definition

Investment are referred to by their round labels such as Seed, Series A, B, C, … late stage, and growth equity.
VC investments excludes debt or other non-equity funding, lending capital, grants and ICOs.

Buyouts, M&A, secondary rounds, and IPOs are treated as exits: excluded from funding data, but included in exit data.

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ENTROPY
INDUSTRIAL
CAPITAL